

Saving by Consuming: The Intertemporal Behavior of Hand-to-Mouth Households^{*}

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- * The experiment developed in this project was registered in the American Economic Association's registry for randomized controlled trials under ID AEARCTR-0012835. The views expressed herein are those of the individual authors and do not necessarily reflect the official positions of the Inter-American Development Bank.

Background

- ▷ Literature on financial constraints & heterogeneous consumption
 - Cornerstone → **Hand-to-Mouth** household, HtM (Campbell and Mankiw, 1989)
 - ↳ Financial exclusion → myopic, **no forward-looking** behavior (Galí et al., 2007)

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 - “*Getting the micro foundations right*” is essential for **aggregate dynamics**
- ▷ **This paper**
 - Revisit the role of the HtM agent by allowing durable consumption
 - ↪ Exhibit forward-looking behavior
 - ↪ Expectations matter for individual & aggregate outcomes

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 - HtM smooths consumption, reallocates spending within-period/across goods
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 - Causally estimate consumption responses to changes in inflation expectations
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3. **Extend PE to GE** → durables in a TANK model
 - Spending becomes more volatile, consumption becomes more stable
 - Durables dampen the amplification of households’ liquidity constraints
 - ↪ As durability increases, TANK & durables → RANK & durables

Contributions

1. Expectations & consumption: Bachmann et al. (2015); Armantier et al. (2016); Cavallo et al. (2017); Vellekoop and Wiederholt (2019); Roth and Wohlfart (2020); Binder and Brunet (2021); D'Acunto et al. (2021); Dräger and Nghiem (2021); Coibion et al. (2022, 2023b,a, 2024); Roth et al. (2023); Andrade et al. (2023); Burke and Ozdagli (2023); Georgarakos et al. (2024); Candia (2024); Hajdini et al. (2024); Jiang et al. (2024); Aktug and Atesagaoglu (2024)
 - Causal evidence linking inflation expectations & consumption on durable categories
 - Effects differ systematically between HtM & non-HtM households
2. The HtM household & TANK models: Campbell and Mankiw (1989); Zeldes (1989); Alvarez et al. (2001); Galí et al. (2007); Bilbiie (2008, 2020); Bilbiie et al. (2012); Kaplan and Violante (2014); Kaplan et al. (2014); Debortoli and Galí (2024); Aguiar et al. (2025), & durable goods: Caballero (1990, 1993); Grossman and Laroque (1990); Eberly (1994); De Gregorio et al. (1998); Leahy and Zeira (2005); Barsky et al. (2007, 2016); Berger and Vavra (2015); McKay et al. (2016); McKay and Wieland (2021, 2022); Beraja and Wolf (2021); Bachmann et al. (2024)
 - Show that durability introduces a forward-looking behavior among HtM households
 - Extend the TANK framework with durables, redefining HtM agents' role in GE models

Two-housheold, two-good framework

Two households & two goods | Key ingredients

- ▷ Two infinitely lived households: $i = \{H, S\}$
 - HtM, $i = H$: financially excluded (Galí et al., 2004; Bilbiie, 2008)
 - non-HtM, $i = S$: full access to financial markets
- ▷ Two goods: $j = \{d, c\}$
 - $c_{i,t}$: non-durable consumption in t
 - $d_{i,t}$: stock of durables at the end of t
- ▷ Durables' law of motion: $d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1}$
 - δ : depreciation rate
 - $z_{i,t}$: new durables purchases in t , reversible
- ▷ One-period, risk-free bond: B_t
 - Only available to non-HtM
 - R_t : gross nominal return, no-arbitrage condition

Households' optimization problem

▷ **HtM** solves

$$\begin{aligned} & \max_{\{z_{i,k}, c_{i,k}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k})] \\ \text{s.t.} \quad & P_{c,t} c_{i,t} + P_{d,t} z_{i,t} = P_{w,t} \\ & d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1} \end{aligned}$$

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▷ **Nomeraire**: non-durable, $P_{c,t}$ = $\begin{cases} q_t = P_{d,t}/P_{c,t} : \text{relative price of durables} \\ w_t = P_{w,t}/P_{c,t} : \text{real wage} \\ r_{t-1} = R_{t-1}/\pi_{c,t} : \text{real interest rate} \end{cases}$

($\pi_{c,t} = P_{c,t}/P_{c,t-1}$: gross inflation)

Durables create an intertemporal trade-off

- ▷ **Intuition:** uncertain future permanent income, two possible states
- **High-income:** $w_k = w_h$ for $k \geq t + 1$, with probability p
 - **Low-income:** $w_k = w_l < w_h$, with probability $(1 - p)$

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- ▷ **Trade-off:** **two benefits offset the cost** of marginally reducing $c_{H,t}$
 - Marginal increase in $d_{H,t} \rightarrow$ immediate utility
 - Undepreciated marginal $d_{H,t} \rightarrow$ discounted expected future utility

$$mu_{H,t}^c = mu_{H,t}^d + \beta (1 - \delta) \{ p \cdot mu_{H,t+1}^d (w_h) + (1 - p) \cdot mu_{H,t+1}^d (w_l) \}$$

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▷ **Durability intertemporal benefits** $\rightarrow$ raise optimal level of $d_{i,t}$

- **Beliefs**/ p shape consumption choices
- Stronger incentives for accumulation when downside risk is high, $w_k = w_l/p = 0$
- Saving vehicle $d_{i,t}$, **smooth consumption** despite lack of bonds

Inflation expectations | Optimal consumption responses

▷ Thought experiment

- One-time/ceteris paribus/exogenous change in expected inflation
- Learned at time t about inflation at $t + 1$, $\mathbb{E}_t [\pi_{j,t+1}]$

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- One-time/ceteris paribus/exogenous change in expected inflation
- Learned at time t about inflation at $t + 1$, $\mathbb{E}_t [\pi_{j,t+1}]$

▷ What are the optimal consumption responses?

$$\frac{\partial z_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \quad \& \quad \frac{\partial c_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \quad \text{for} \quad \begin{cases} \text{household } i \in \{H, S\} \\ \text{price } j \in \{d, c, w\} \end{cases}$$

- Policy functions differ by household type
- Responses operate through expected relative prices: $\{q_t, w_t, r_{t-1}\}$
- HtM are insensitive to changes in expected real interest rate

No durables, $\delta = 1$ | Myopic HtM

Proposition 1

If durables fully depreciate ($\delta = 1$), the HtM household is unresponsive to changes in inflation expectations

$$\left. \frac{\partial z_{H,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \right|_{\delta=1} = \left. \frac{\partial c_{H,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \right|_{\delta=1} = 0$$

By contrast, the non-HtM household remains responsive even when $\delta = 1$, adjusting consumption of all goods in the same direction.

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By contrast, the non-HtM household remains responsive even when $\delta = 1$, adjusting consumption of all goods in the same direction.

- ▷ No durables $\rightarrow$ HtM expectations play no role
- ▷ non-HtM expectations still matter $\rightarrow$ intertemporal substitution via financial markets

With durables, $\delta < 1$ | Expected wage inflation shock

Proposition 2

Given a one-time/ceteris paribus/exogenous change in the expected inflation of the nominal wage $\mathbb{E}_t [\pi_{w,t+1}]$, the HtM responses satisfy

$$\frac{\partial z_{H,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} < 0 \quad \& \quad \frac{\partial c_{H,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0$$

By contrast, given the non-arbitrage condition, the non-HtM responses satisfy

$$\frac{\partial z_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0 \quad \& \quad \frac{\partial c_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} > 0 \quad \& \quad \frac{\partial b_{S,t}}{\partial \mathbb{E}_t [\pi_{w,t+1}]} < 0$$

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▷ Incentives to pull future resources forward

- non-HtM raises overall consumption, reducing net savings
- Financial friction → HtM reallocates fixed resources within period

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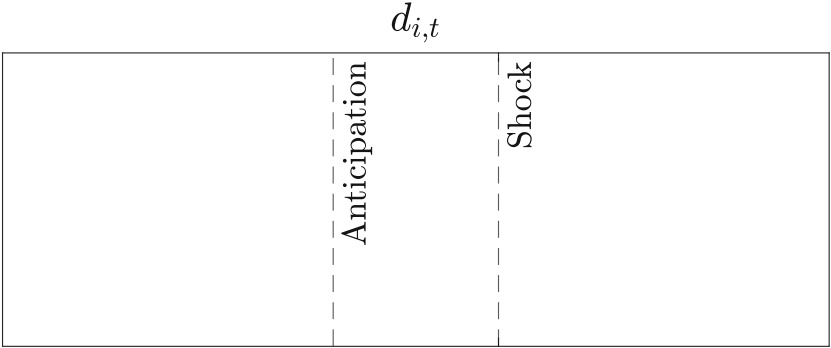
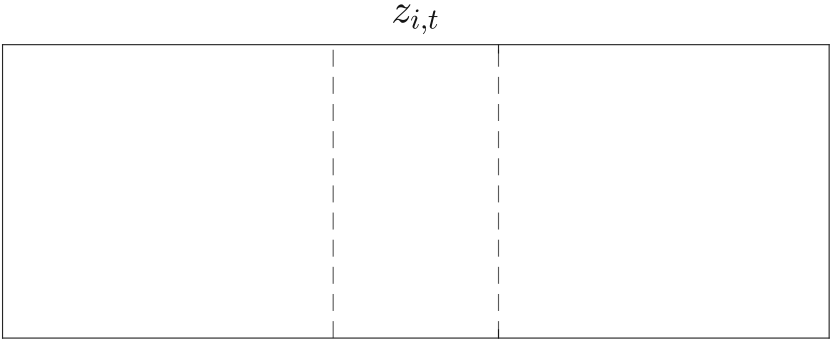
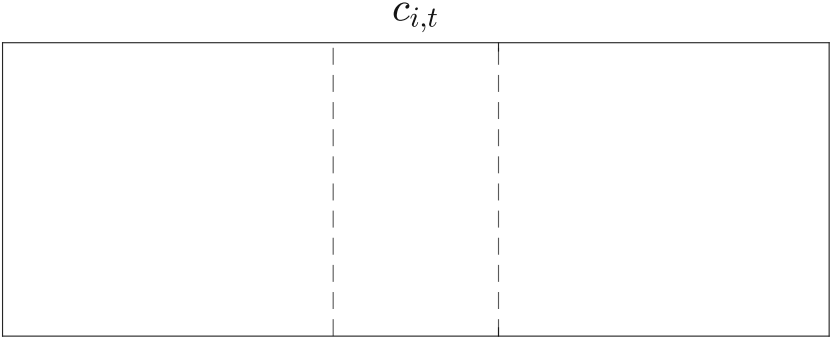
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- ▷ Incentives to pull future resources forward
 - **non-HtM raises overall consumption**, reducing net savings
 - Financial friction → **HtM reallocates fixed resources** within period
- ▷ To smooth consumption, HtM use **durables** as a “*borrowing from future self*” instrument

Expected wage inflation shock

▷ Decline in expected wage inflation



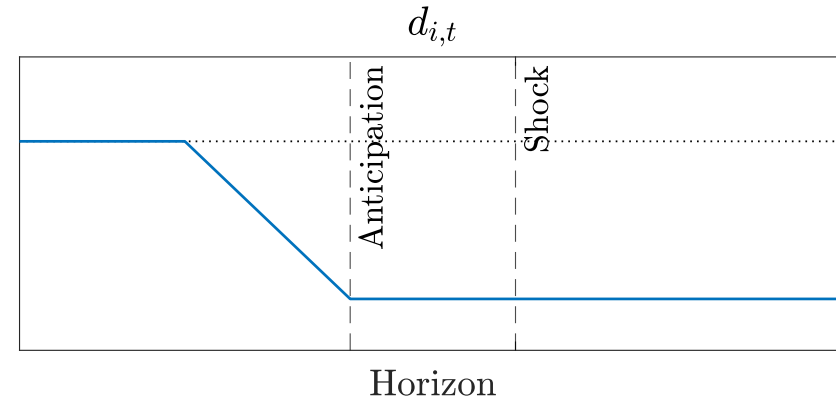
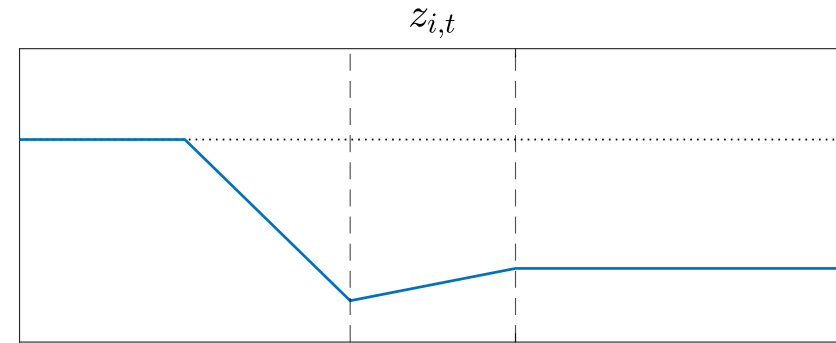
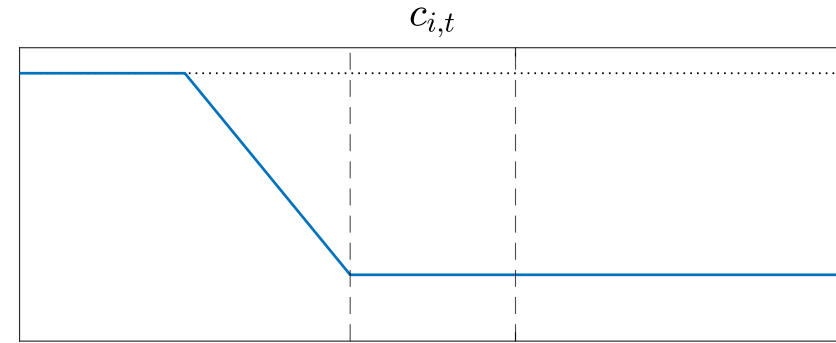
Horizon

— Non-HtM
- - HtM
- . HtM ($\delta = 1$)

Expected wage inflation shock

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- **non-HtM**: Adjusts according to the **PIH**

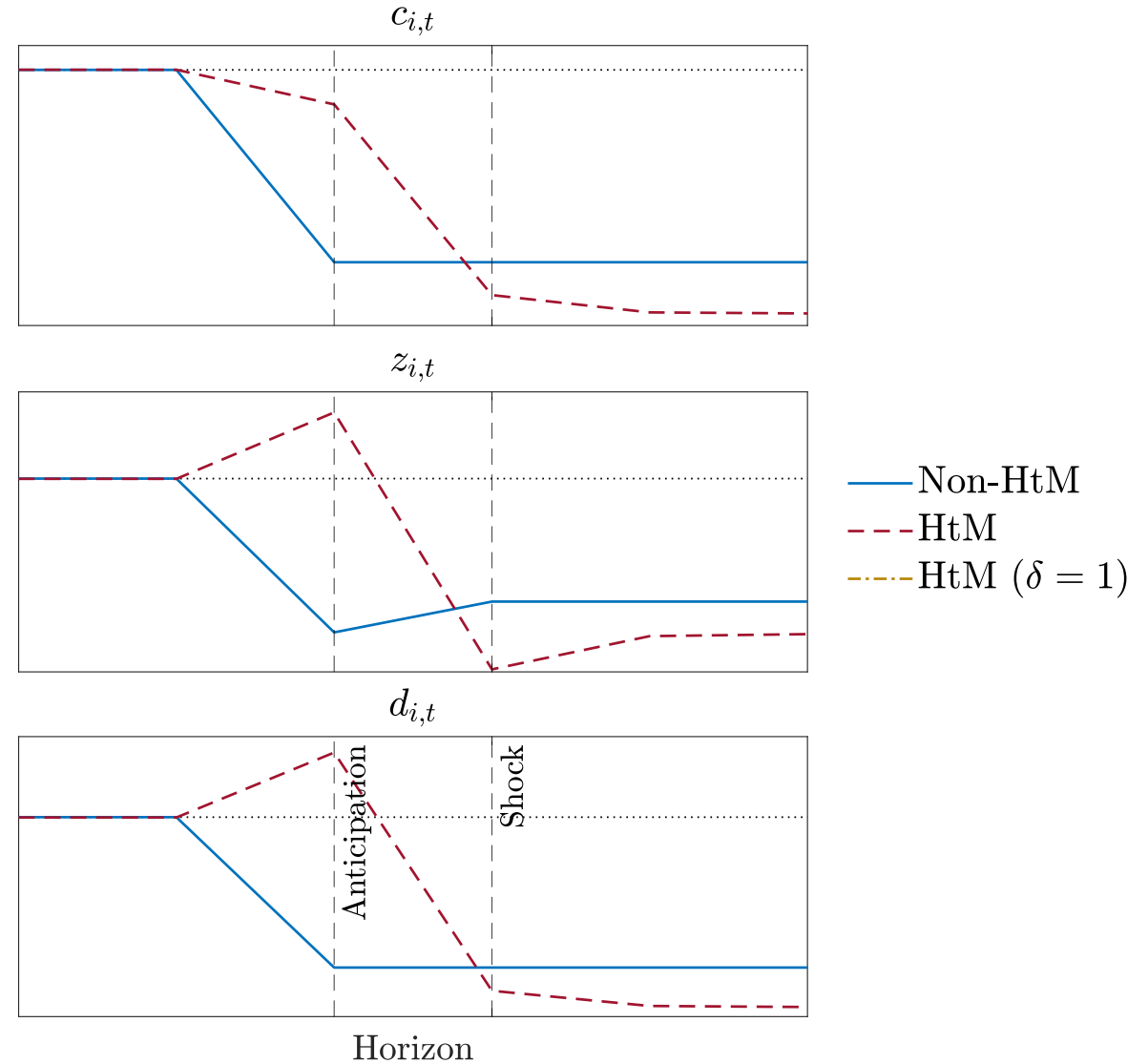


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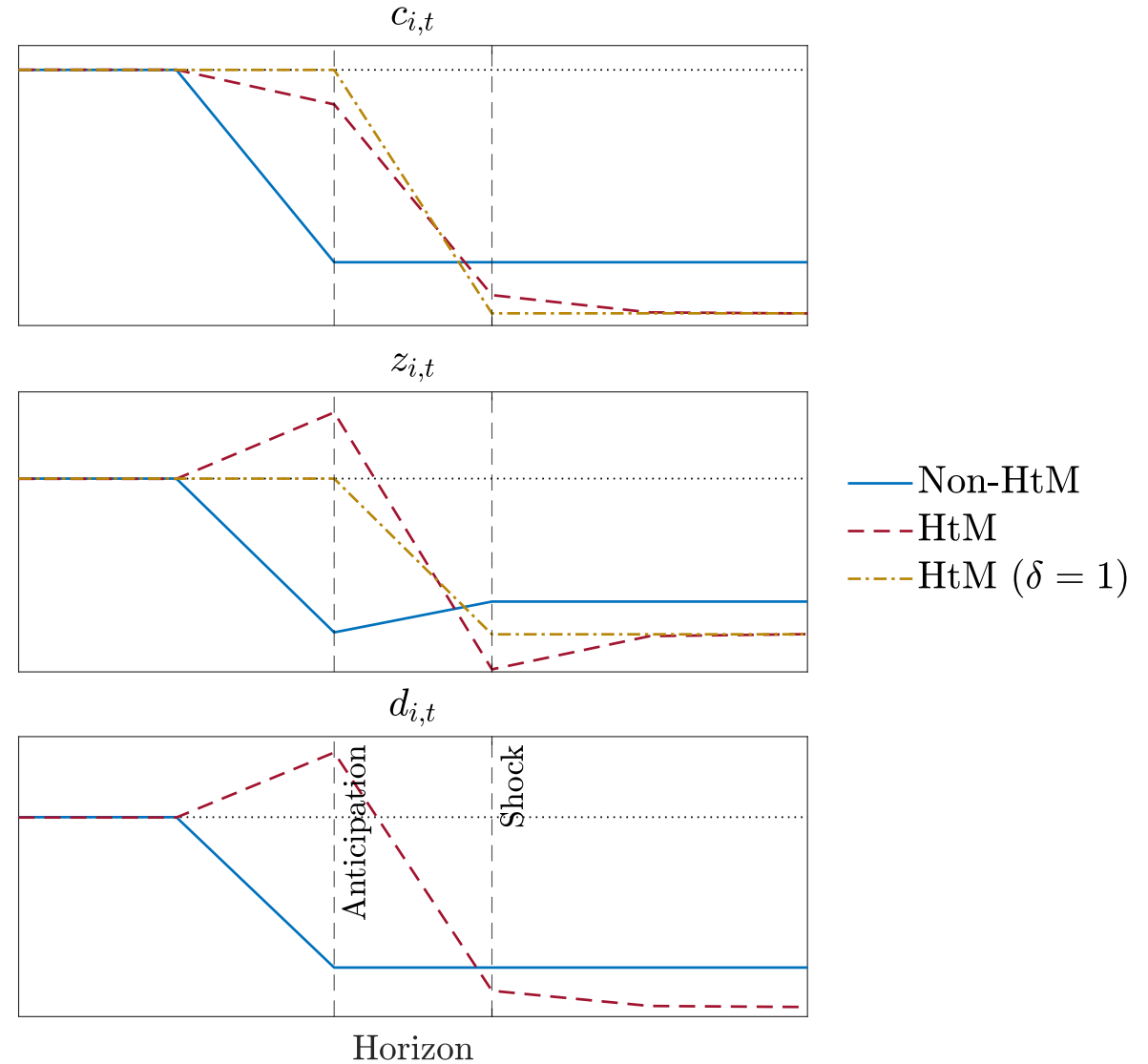
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Expected wage inflation shock

▷ Decline in expected wage inflation

- **non-HtM**: Adjusts according to the PIH
- **Forward-looking HtM**: Accelerates durable accumulation, saving vehicle
- **Myopic HtM** ($\delta = 1$): Remains unresponsive



Inflation expectations shocks | Predictions

▷ **Prediction 1:** ↑ **aggregate** inflation expectations

Household	Higher aggregate inflation expectations		
	Durable	Non-durable	
HtM			
non-HtM			

Inflation expectations shocks | Predictions

▷ **Prediction 1:** ↑ aggregate inflation expectations = ↓ expected real wages (Hajdini et al., 2022)

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non-HtM	Up/Down	Up/Down	

Inflation expectations shocks | Predictions

- ▷ **Prediction 1:** ↑ **aggregate** inflation expectations = ↓ expected **real wages** & ↓ **real rate**
- ▷ **Prediction 2:** ↑ **durable** inflation expectations = ↑ expected **relative price of durables**

Household	Higher aggregate inflation expectations		Higher durable inflation expectations	
	Durable	Non-durable	Durable	Non-durable
HtM	Up	Down	Up	Down
non-HtM	Up/Down	Up/Down	Up	Up/Down

Data

Data source | Cleveland Fed's Survey of Consumer Expectations

▷ Taking the theory to data $\Rightarrow$ empirical challenge

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- ▷ Taking the theory to data $\Rightarrow$ empirical challenge $\Rightarrow$ **randomized controlled trial**
 - **Clean & controlled environment** designed to evaluate predictions
 - **Treatments** $\rightarrow$ **exogenous variation** in expectations

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- ▷ Taking the theory to data $\Rightarrow$ empirical challenge $\Rightarrow$ **randomized controlled trial**
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 - Designed by us for this experiment
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 - Implemented in the Cleveland Fed's SCE (Knotek et al., 2020)
- ▷ Key survey features
 - Repeated cross-section with daily frequency
 - Online, US-representative sample restricted to +18 years old
- ▷ Our sample
 - $\approx$ 5,500 valid observations after standard filters (Haaland et al., 2023)
 - Fielded between November 21, 2023, & February 13, 2024

Experimental setup | Timeline

I. Baseline information: (1) background → **HtM status**, (2) past consumption expenditure, & (3) short-run inflation expectations → **prior**

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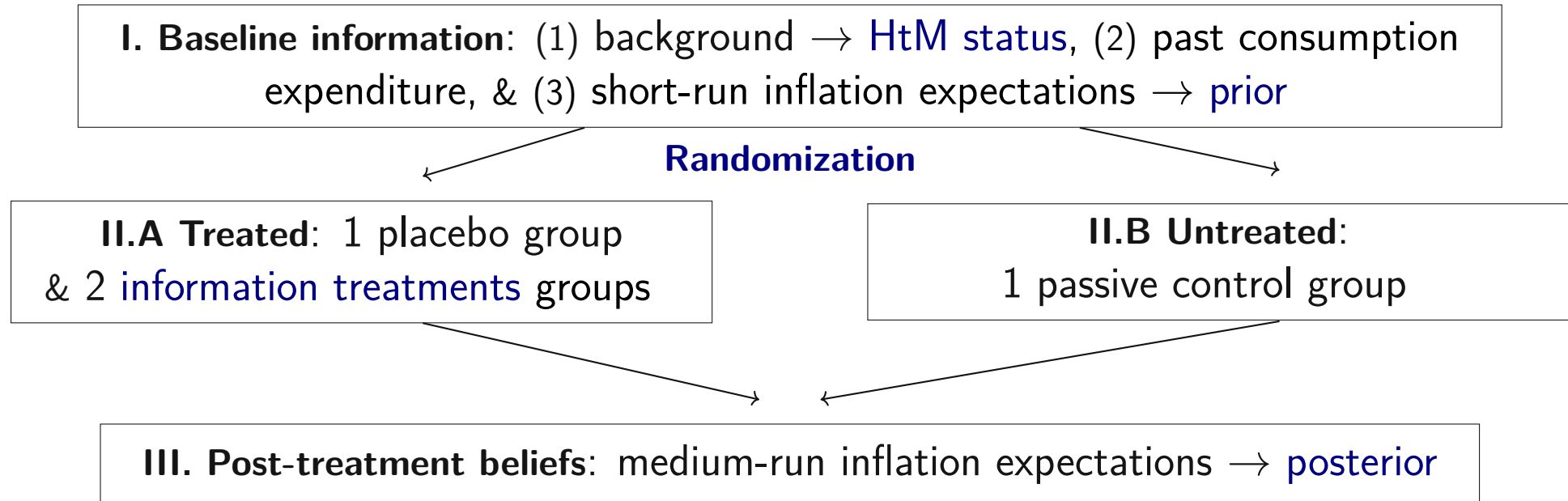
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Randomization

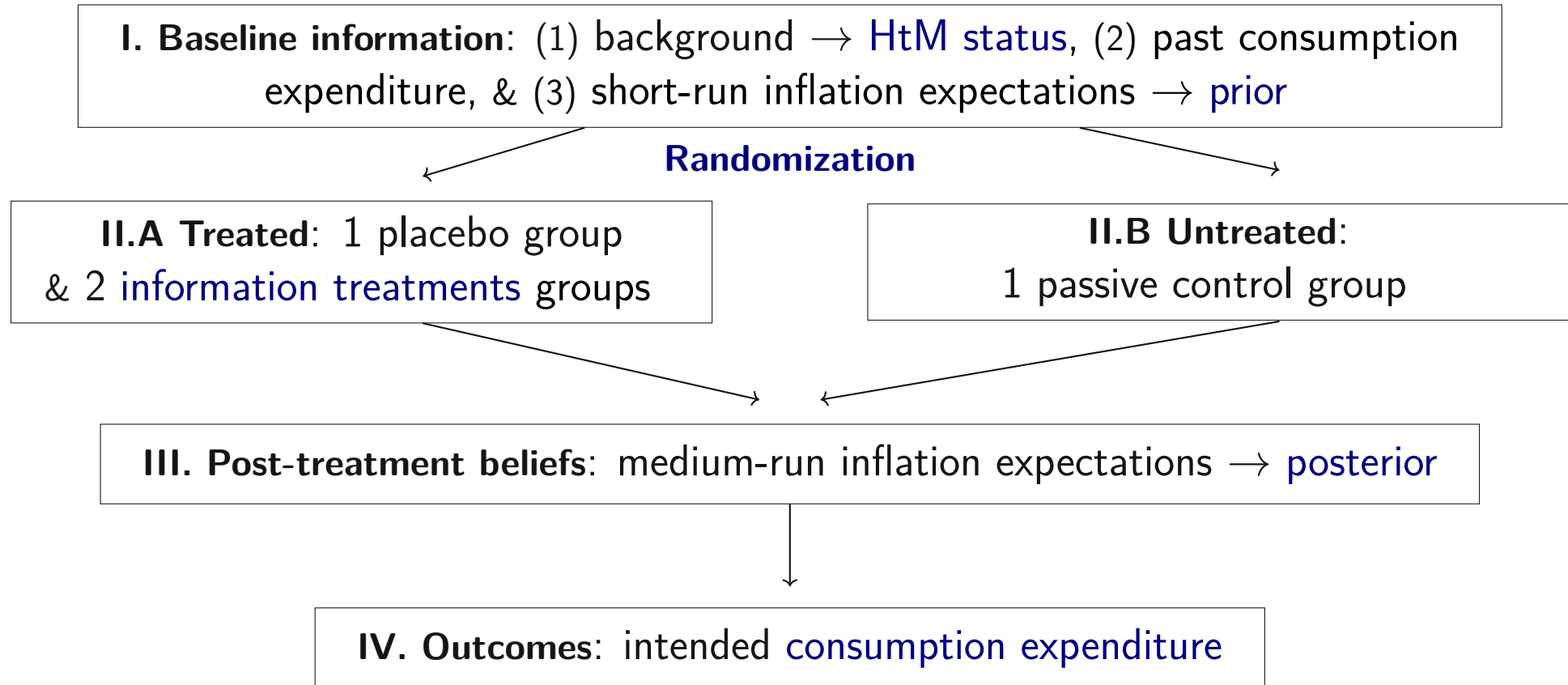
II.A Treated: 1 placebo group
& 2 **information treatments** groups

II.B Untreated:
1 passive control group

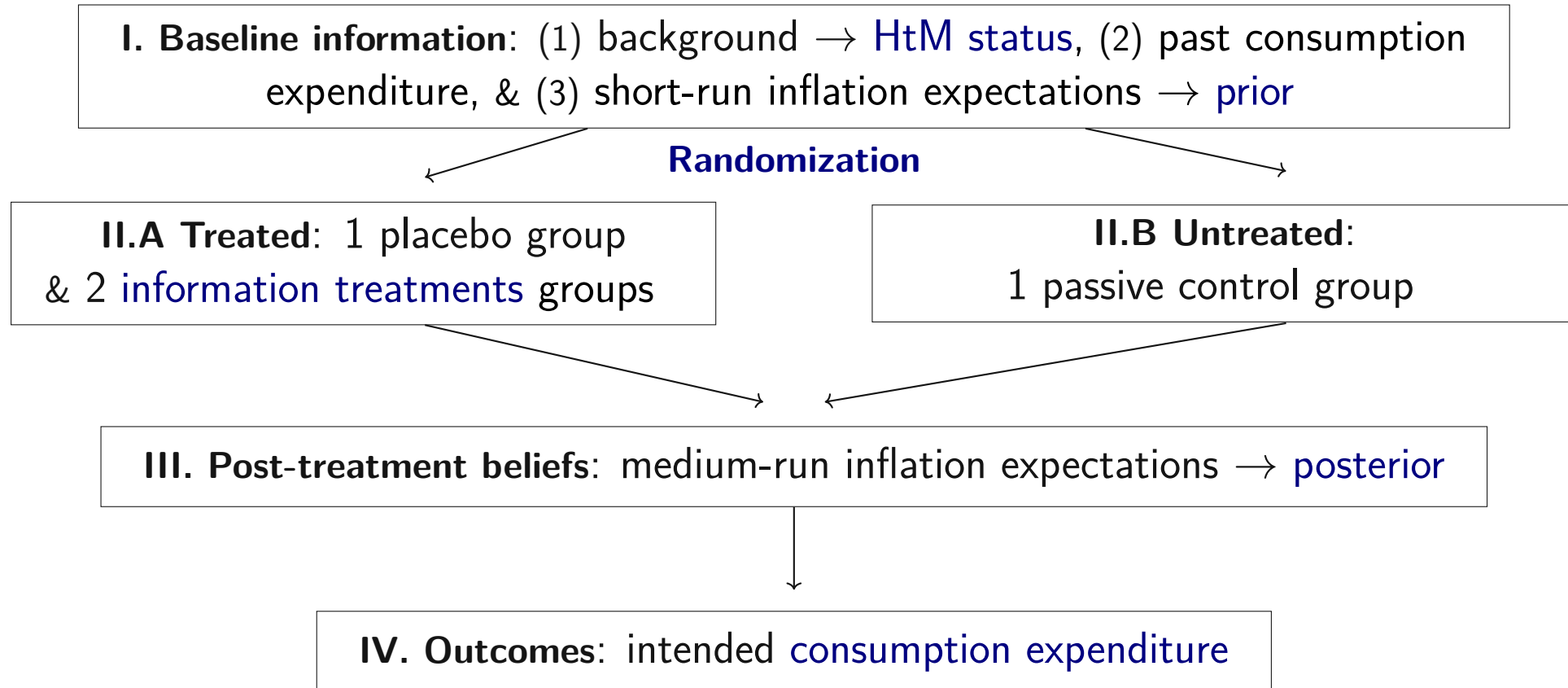
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▷ **Randomization** → isolates the **causal effect** of inflation expectations on consumption

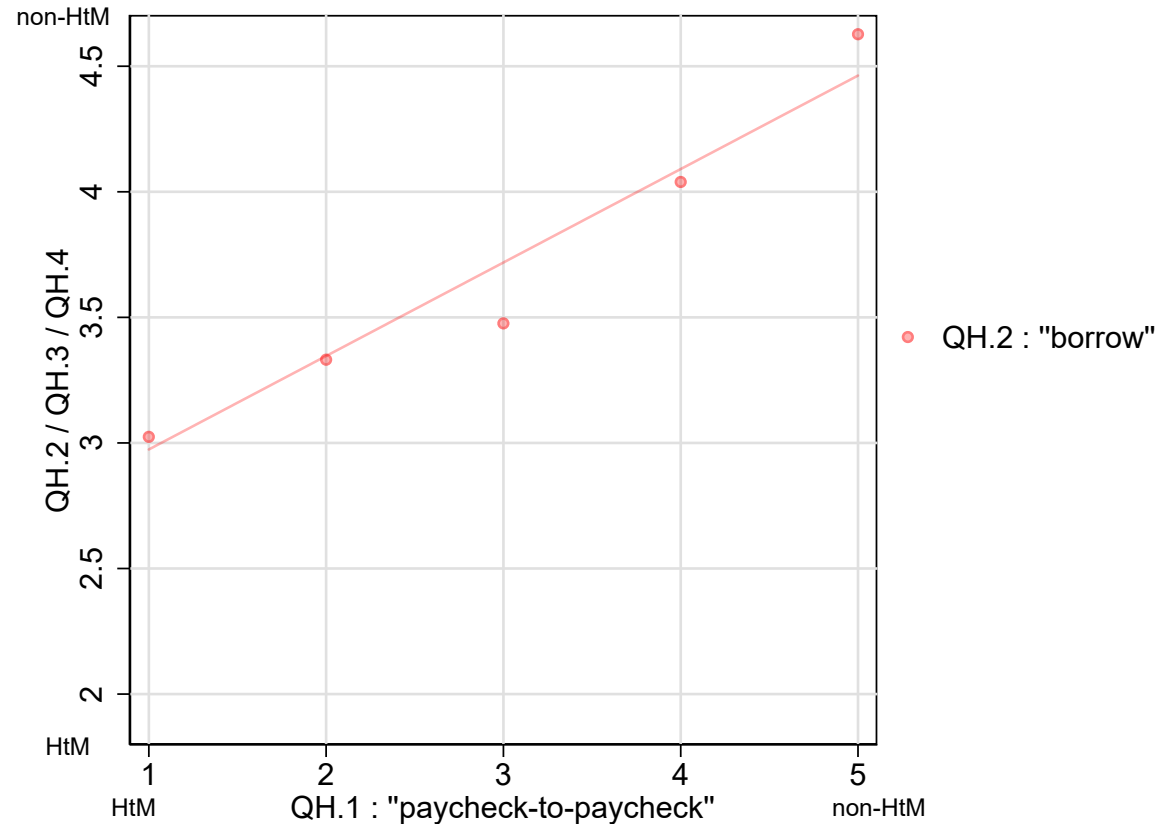
HtM candidates | Who are the HtM?

- ▷ Identifying HtM respondents is challenging/costly
- ▷ Our approach: **key indicators of HtM status**
 - Self-perception of “*living paycheck-to-paycheck*”
 - Perceived **financial constraints**
 - Typical **saving behavior**
 - **Liquid assets** relative to typical consumption

▶ **HtM questions**

HtM measures | Strong internal consistency

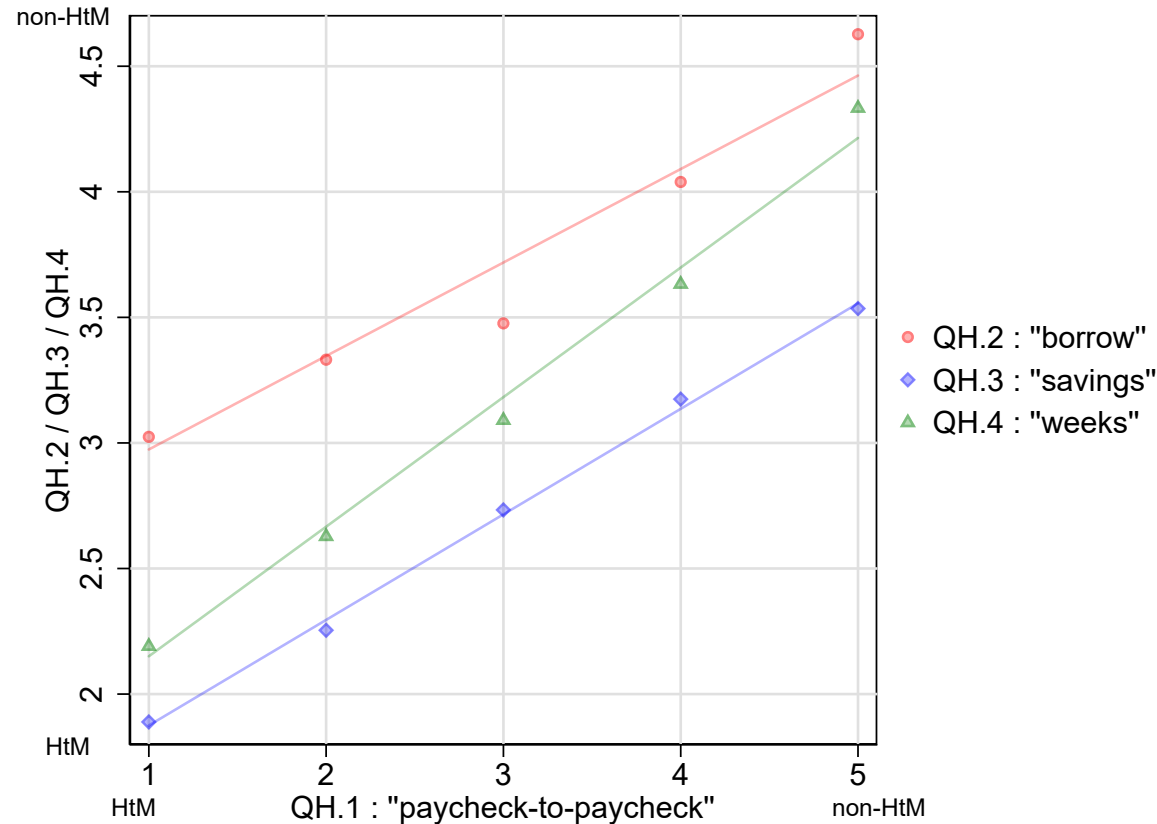
Relationship between the answers to the different HtM questions



- Lower recoded values → higher probability of being HtM
- Strong individual-level correlation → consistent classification across measures

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▶ HtM questions

- ▷ Dimension reduction: PCA synthesizes multivariate information → PC1 (single measure)

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 - **Liquid assets** relative to typical consumption
- ▷ Dimension reduction: PCA synthesizes multivariate information → **PC1** (single measure)
 - PC1 captures a **dominant & significant** share of common variation, $\approx 60\%$
 - PC1 eigenvalue $\gg$ Kaiser threshold

▶ **HtM questions**

▶ **Scree plot**

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- Typical **saving behavior**
- **Liquid assets** relative to typical consumption

▶ **HtM questions**

▷ Dimension reduction: PCA synthesizes multivariate information → **PC1** (single measure)

- PC1 captures a **dominant & significant** share of common variation, $\approx 60\%$
- PC1 eigenvalue $\gg$ Kaiser threshold

▶ **Scree plot**

▷ **HtM: baseline** → **bottom 40%** of PC1 distribution; **robustness** → **30%-50%** (Campbell and Mankiw, 1989; Kaplan and Violante, 2014; Kaplan et al., 2014; Aguiar et al., 2025)

HtM candidates | Who are the HtM?

Demographic & socioeconomic variables by HtM status

	Full sample	non-HtM	HtM
Age	49.1	49.8	48.1
Female	0.50	0.44	0.58
Number of children	1.50	1.55	1.44
Education	Some college, but no degree	Some college, but no degree	Some college, but no degree
Income	\$35,000-\$49,999	\$50,000-\$99,999	\$35,000-\$49,999
Last month expenditure, "Aggregate" option	\$2,744	\$2,850	\$2,600
Completion time	12.68	12.60	12.80
Civil status, "Married" option	48.8%	58.7%	34.7%
Political affiliation, "Democrat" option	40.2%	44.3%	34.3%
Observations	5,529	3,231	2,298

▷ **HtM** → lower-income, more likely to be single or divorced & to identify as female

Inflation expectations | Prior beliefs

- ▷ Quantitative point forecast → accurate first-moment predictions (Clements, 2010, 2014)
- ▷ Two-step elicitation → lower measurement error (Armantier et al., 2017)
- ▷ **Outcome** → clean, low-noise measure of short-run inflation expectations

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Prior : *short-run inflation expectations*

Q1.1 Over the next 12 months, do you think that there will be inflation or deflation?

- ☐ Inflation
- ☐ Deflation (the opposite of inflation)

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- ☐ Deflation (the opposite of inflation)

As a follow-up question...

↪ What do you expect the rate of <inflation> <deflation> to be over the next 12 months?

Please give your best guess.

- I expect the rate of <inflation> <deflation> to be _____ percent over the next 12 months.

Information treatments | Random assignment

- ▷ Truthful, official, & publicly available information → attend to & trust the information

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Treatment 1 : *shift aggregate inflation expectations*

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Treatment 1 : *shift aggregate inflation expectations*

- T1 *According to the Survey of Professional Forecasters conducted by the [Federal Reserve Bank of Philadelphia](#), the annual aggregate inflation rate at the end of the year 2024 will be 2.5%. / 1,410 observations*

Treatment 2 : *shift durables relative price*

- T2 *According to the [Bureau of Labor Statistics](#), between 2012 and 2019, average apparel prices stayed relatively constant. On average, these prices decreased by 0.24% annually. In the last 12 months, average apparel prices have increased by 3.1%. / 1,354 observations*

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Placebo : *past population growth*

P According to the *Census*, the U.S. population grew 2.9% between 2018 and 2022. / 1,370 observations

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P According to the *Census*, the U.S. population grew 2.9% between 2018 and 2022. / 1,370 observations

C **Passive control group** → no information provided, benchmark / 1,395 observations

► Balance tests

Inflation expectations | Posterior beliefs

▷ Different horizon → avoids question repetition bias (Stantcheva, 2022)

Inflation expectations | Posterior beliefs

▷ Different horizon → avoids question repetition bias (Stantcheva, 2022)

Posterior : *medium-run inflation expectations*

Q1.2 Over the next 5 years, do you think there will be inflation or deflation on average?

- ☐ Inflation
- ☐ Deflation (the opposite of inflation)

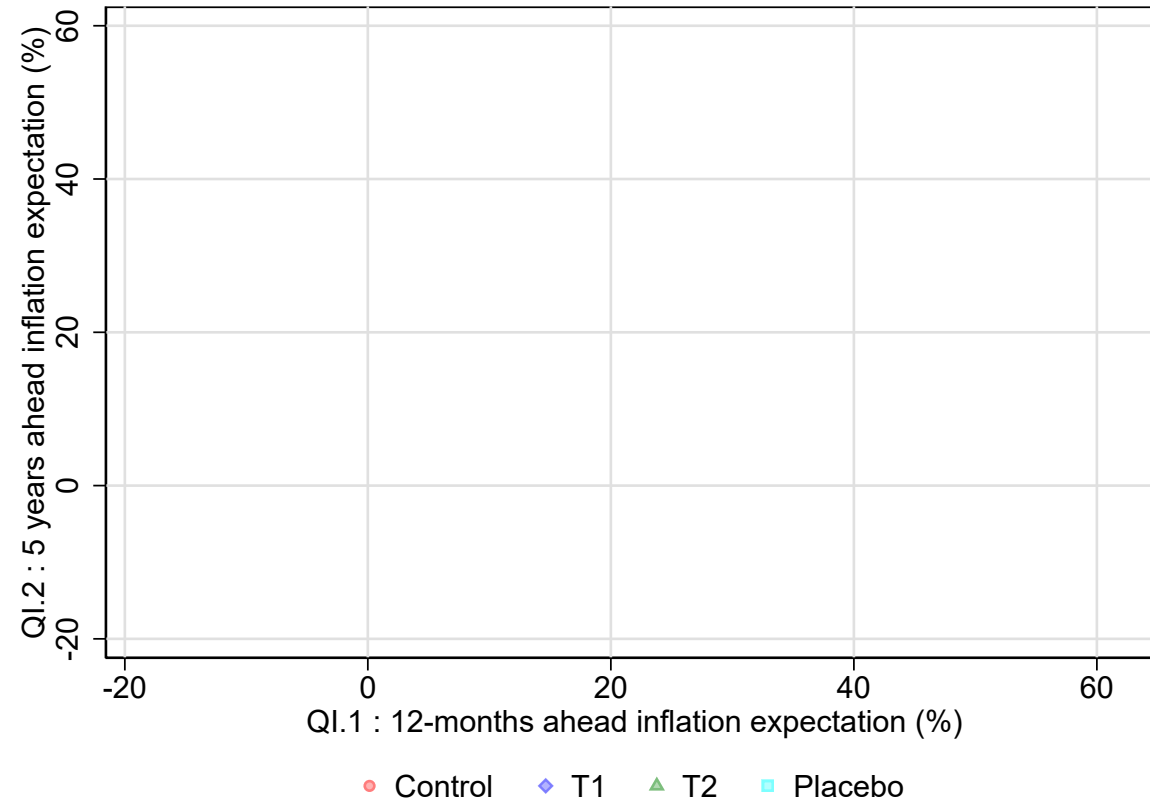
As a follow-up question...

↪ What do you expect the average annual rate of $\langle \text{inflation} \rangle$ $\langle \text{deflation} \rangle$ to be over the next 5 years? Please give your best guess.

- I expect the average annual rate of $\langle \text{inflation} \rangle$ $\langle \text{deflation} \rangle$ to be _____ percent per year over the next 5 years.

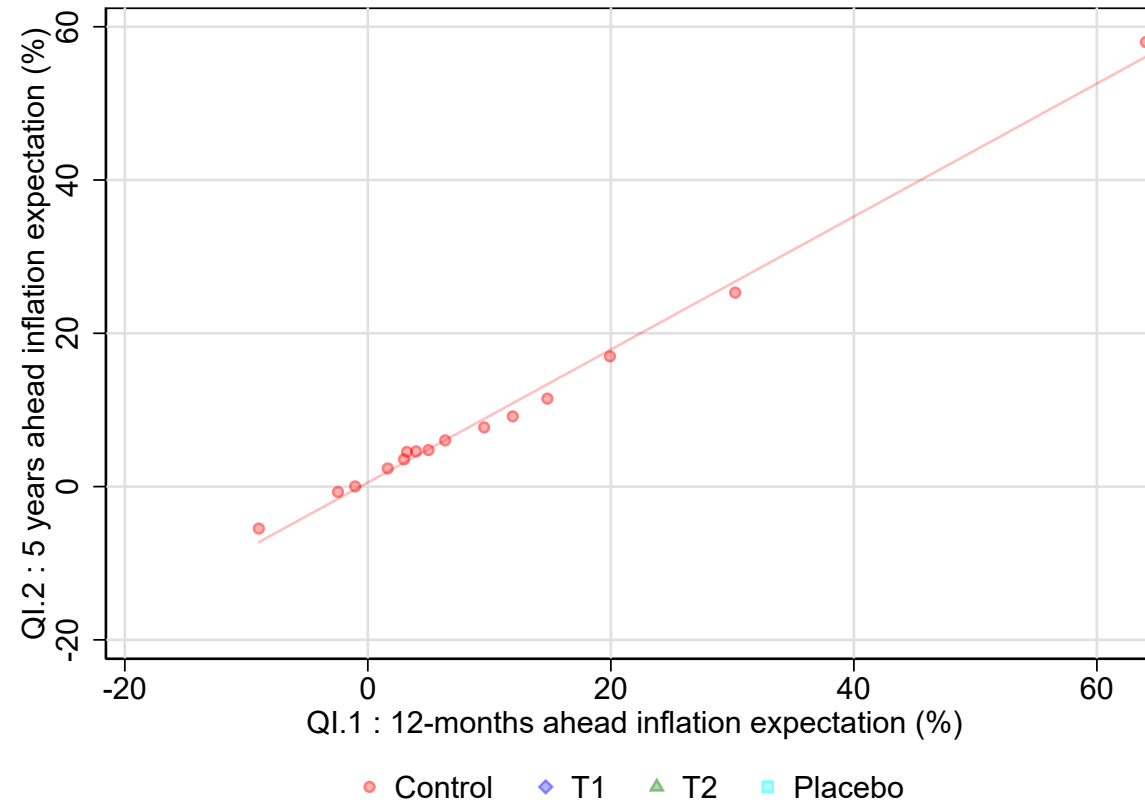
Treatments | Beliefs updating, first-stage

Prior v/s posterior beliefs by treatment status



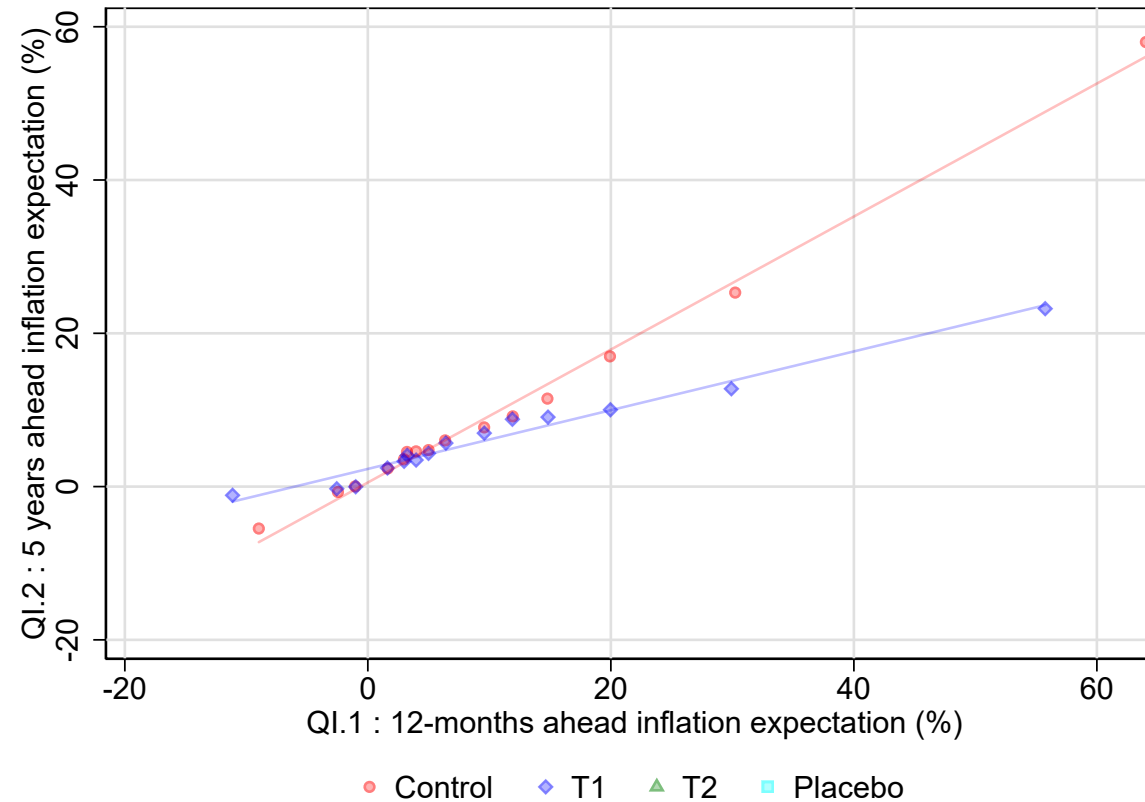
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Prior v/s posterior beliefs by treatment status



Treatments | Beliefs updating, first-stage

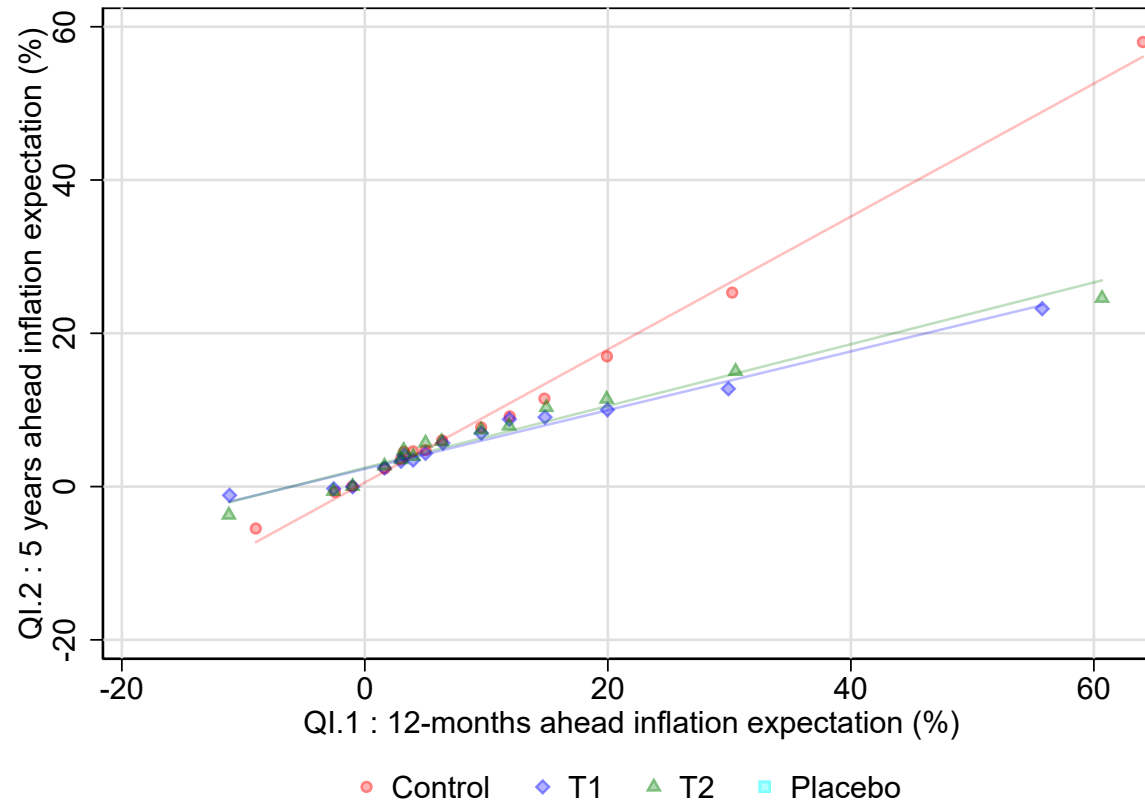
Prior v/s posterior beliefs by treatment status



- Treatments moderate expectations

Treatments | Beliefs updating, first-stage

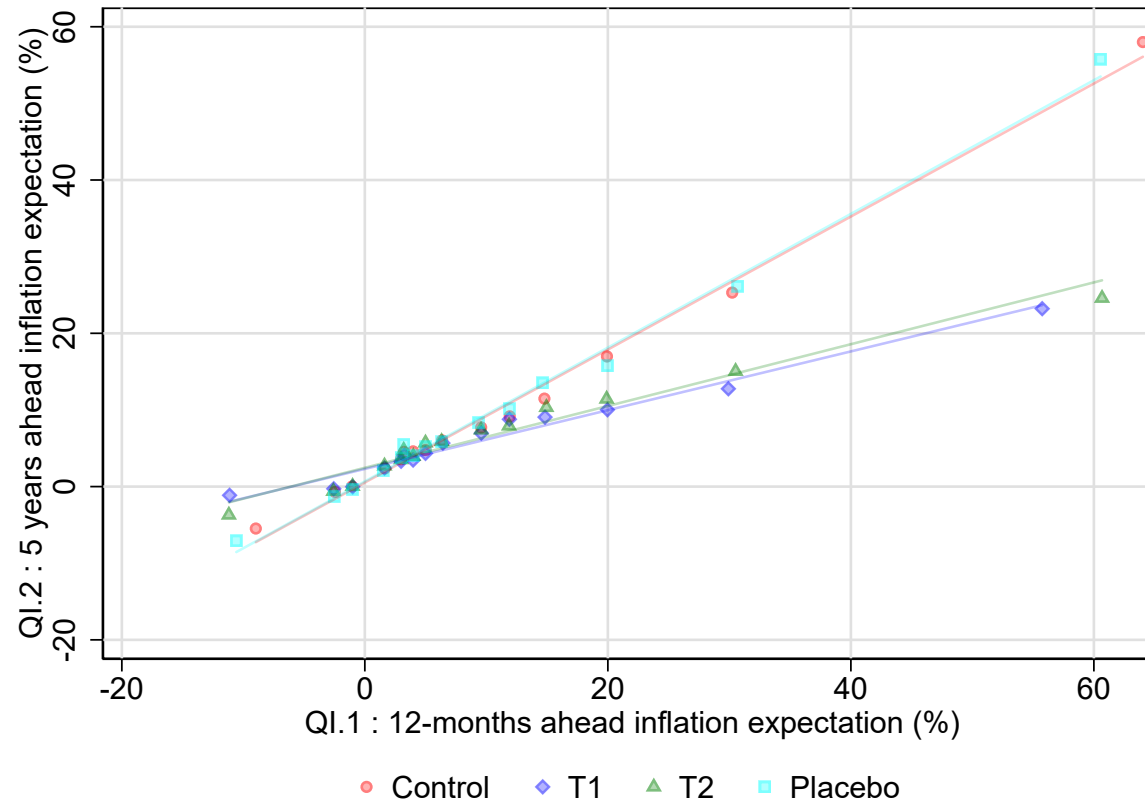
Prior v/s posterior beliefs by treatment status



- Treatments moderate expectations, similarly across treated groups

Treatments | Beliefs updating, first-stage

Prior v/s posterior beliefs by treatment status



- Treatments moderate expectations, similarly across treated groups
- Placebo indistinguishable from control group

Treatments | Beliefs updating, first-stage

▷ To evaluate treatment effects on medium-run expectations

$$\textcolor{brown}{post}_j = \alpha + \mathbf{X}_j \boldsymbol{\beta}' + \eta \cdot \textcolor{brown}{prior}_j + \sum_{\tau=\{1,2,P\}} [\gamma_{\tau,1} \cdot \textcolor{teal}{treat}_{j,\tau} + \gamma_{\tau,2} (\textcolor{teal}{treat}_{j,\tau} \times \textcolor{brown}{prior}_j)] + v_j$$

- $\textcolor{brown}{post}_j$: 5-year average annual inflation expectations of respondent j , posterior
- $\textcolor{brown}{prior}_j$: 12-month inflation expectations, prior
- $\textcolor{teal}{treat}_{j,\tau}$: 1 if j received the treatment $\tau = \{1, 2\}$ or placebo P, & 0 otherwise
- $\mathbf{X}_j$: baseline respondent-level covariates, measured before treatment implementation

Treatments | Beliefs updating, first-stage

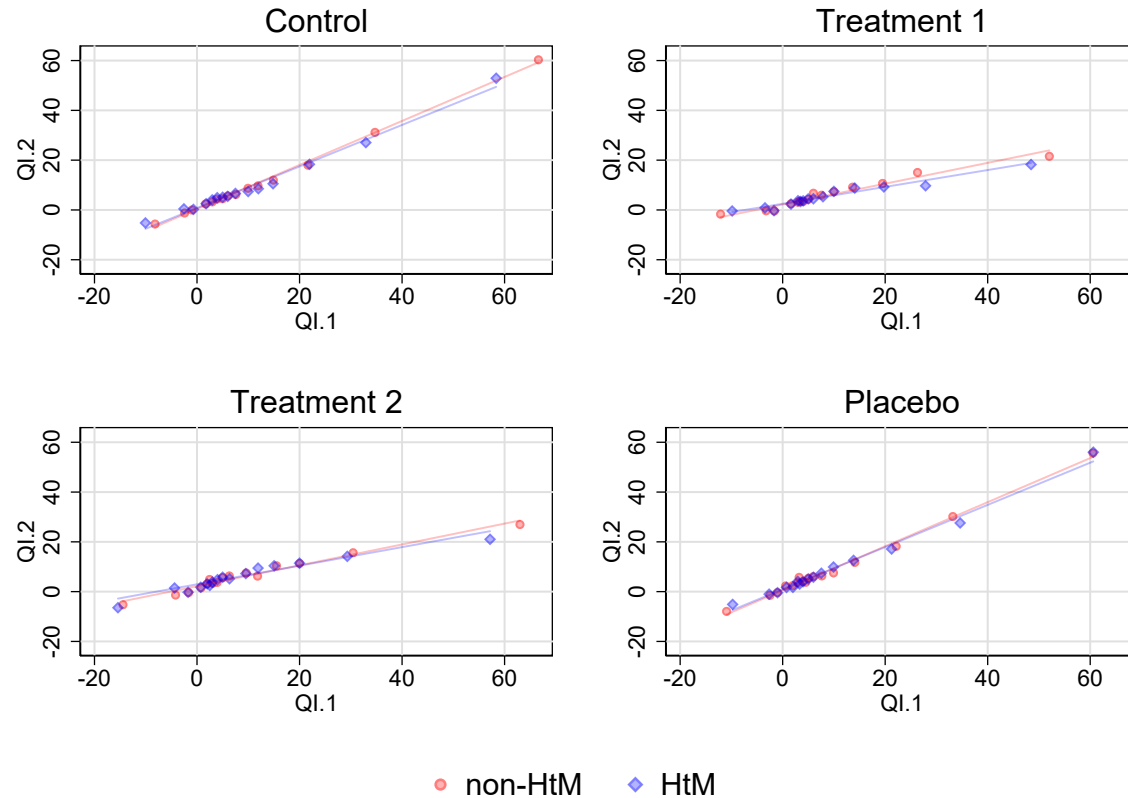
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- $\mathbf{X}_j$: baseline respondent-level covariates, measured before treatment implementation
- **Key coefficients:** $\gamma_{\tau,2}$ captures learning from treatment τ
- **Random assignment** $\rightarrow \gamma_{\tau,2}$ identifies causal learning from information

Treatments across HtM status | Validation

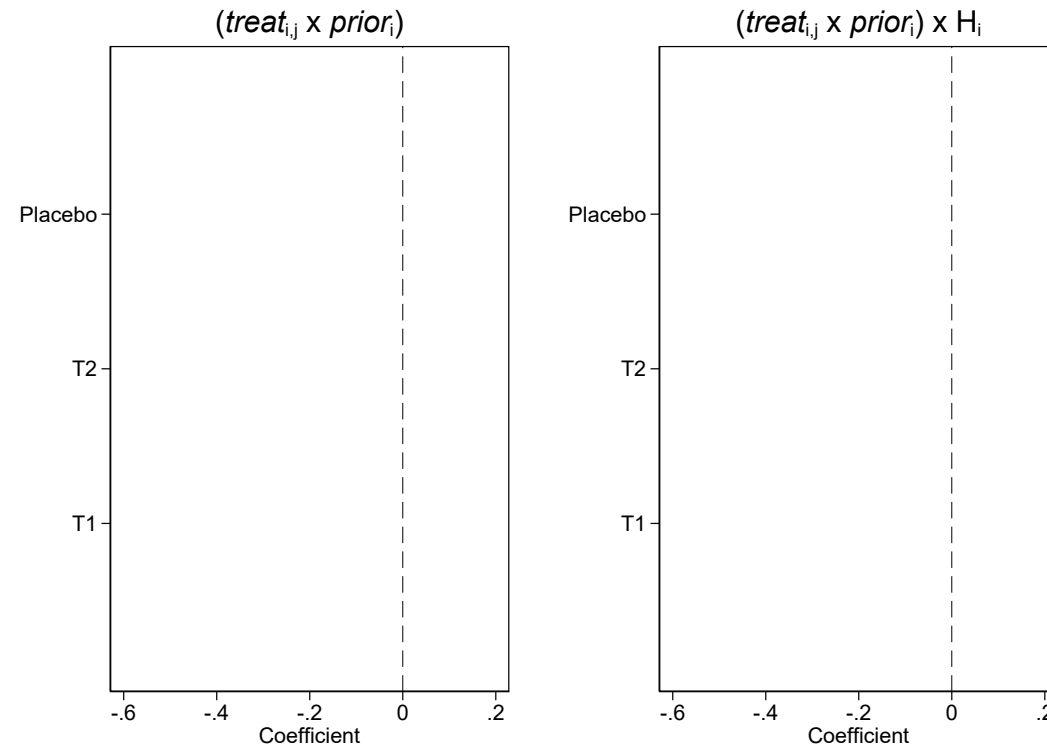
Prior v/s posterior beliefs by treatment & HtM status



- HtM & non-HtM respondents exhibit **similar slopes** across treatment groups
- Suggests **no significant differences in beliefs updating** by HtM status

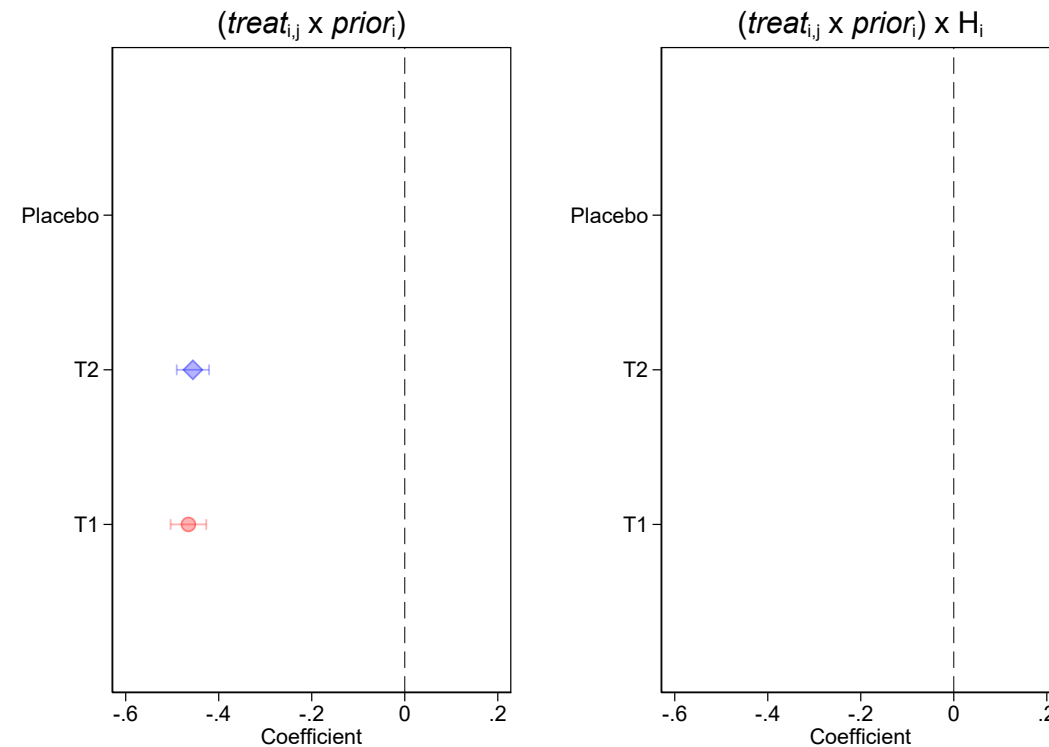
Treatments | Three key results on belief updating

Treatment effects on medium-run inflation expectations



Treatments | Three key results on belief updating

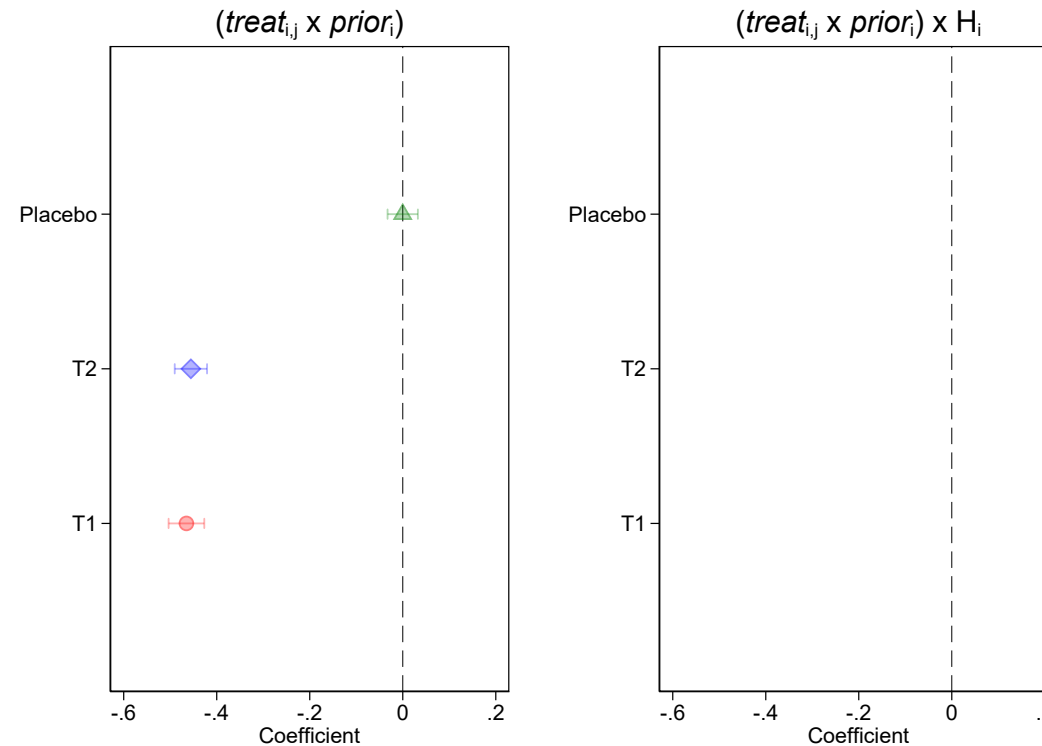
Treatment effects on medium-run inflation expectations



- T1 & T2 → significant revision of expectations

Treatments | Three key results on belief updating

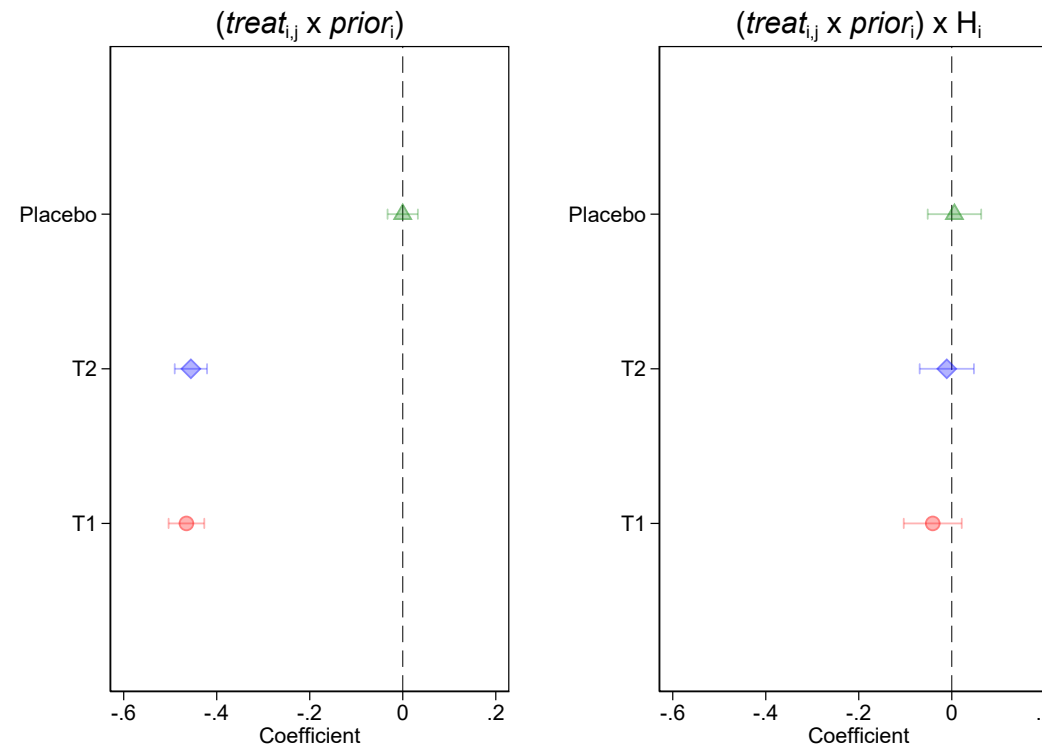
Treatment effects on medium-run inflation expectations



- T1 & T2 → significant revision of expectations
- Placebo → no belief updating, no numerical anchoring bias

Treatments | Three key results on belief updating

Treatment effects on medium-run inflation expectations



- T1 & T2 → significant revision of expectations
- Placebo → no belief updating, no numerical anchoring bias
- HtM v/s non-HtM → no differential treatment assimilation

Past & intended consumption | Survey questions

- ▷ Survey spending plans track realized behavior (Schnorpfeil et al., 2023; Kotsogiannis and Sakellaris, 2025)
- ▷ Overall spending & breakdown into four consumption categories (Bank of Canada)

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Intended consumption : *total spending*

QC.1 Compared with your spending last month, how do you expect your total spending to change in the next ... 1. One month? / 2. One year?

- ☐ Go up by _____ %.
- ☐ No change.
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- **Services:** *medical and dental care, haircuts, and restaurant meals...*
- **Non-durables:** *food and beverage products from grocery stores, and gasoline and other energy goods...*
- **Semi-durables:** *clothing and footwear, jewelry, watches, silverware, toys, tools and garden equipment, and household textiles and utensils...*
- **Durables:** *household appliances, radio and television sets, sporting and wheeled goods...*

Empirical strategy & results

Identification strategy, no heterogeneity | 2SLS

▷ First-stage → **expectations updating**

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▷ First-stage → **expectations updating**

$$post_j = \alpha + \mathbf{X}_j \beta' + \eta \cdot prior_j + \sum_{\tau=\{1,2\}} [\gamma_{\tau,1} \cdot treat_{j,\tau} + \gamma_{\tau,2} (treat_{j,\tau} \times prior_j)] + v_j$$

- $post_j$: 5-year average annual inflation expectations of respondent j
- $prior_j$: 12-month inflation expectations
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▷ Second-stage → **consumption responses**

$$\% \Delta c_{j,k,h} = \alpha_{k,h} + \mathbf{X}_j \beta'_{k,h} + \eta_{k,h} \cdot \text{prior}_j + \theta_{k,h} \cdot \widehat{\text{post}}_j + \varepsilon_{j,k,h}$$

- $\% \Delta c_{j,k,h}$: respondent j 's intended expenditure growth on category k at horizon h
- **Key coefficients:** $\theta_{k,h} \rightarrow$ expectations' **causal effect** on $\% \Delta c_{j,k,h}$

Heterogeneous effects by HtM status | 2SLS

▷ Adapt the first stage to allow for heterogeneity

- HtM identifier
$$H_j = \begin{cases} 1 & \text{if respondent } j \text{ is HtM} \\ 0 & \text{otherwise} \end{cases}$$
- Two household-type-specific variables to instrumentalize

$$[\textit{post}_j \times (1 - H_j) , \textit{post}_j \times H_j]$$

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▷ Second-stage → **heterogeneous consumption responses**

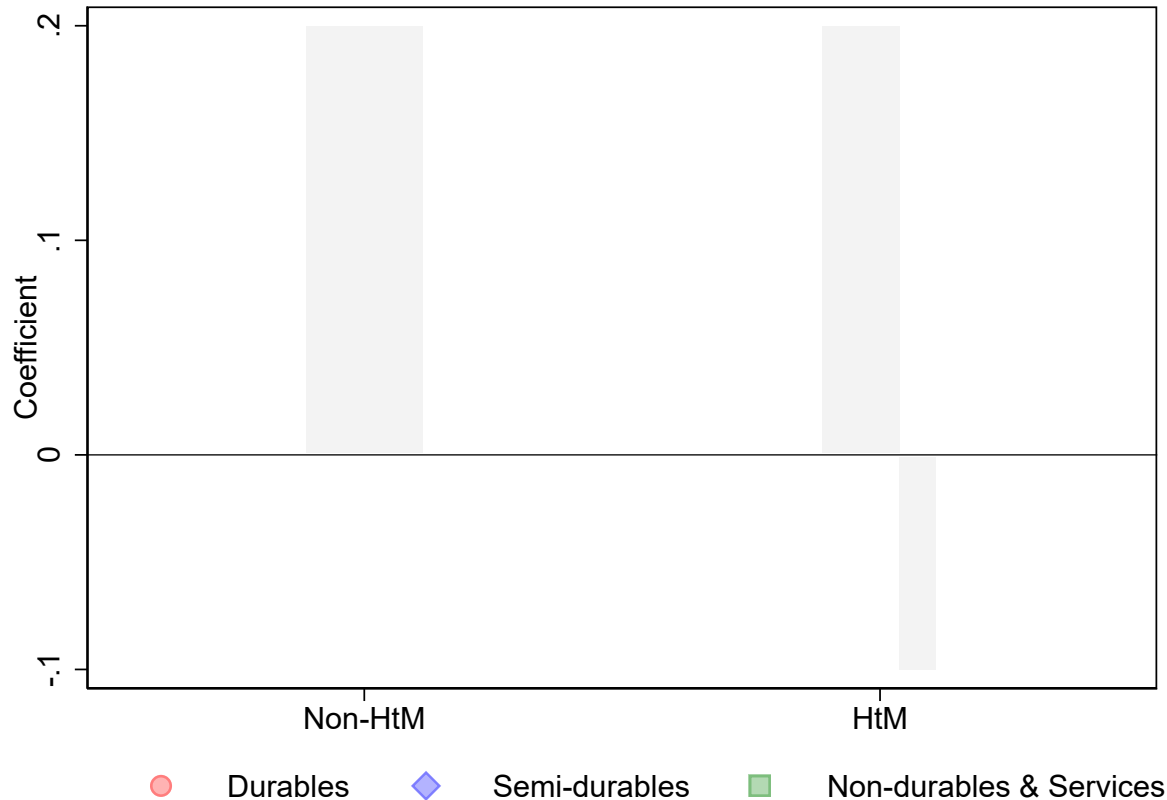
$$\% \Delta c_{j,k,h} = \alpha_{k,h} + \mathbf{X}_j \boldsymbol{\beta}'_{k,h} + \eta_{k,h} \cdot \textit{prior}_j + \theta_{k,h}^N \cdot \overbrace{\textit{post}_j \times (1 - H_j)} + \theta_{k,h}^H \cdot \overbrace{\textit{post}_j \times H_j} + \varepsilon_{j,k,h}$$

- **Key coefficients:** $\theta_{k,h}^i \rightarrow$ causal effect of expectations for $i = \{H, N\}$

► Regressions details

PE | Prediction 1: Higher aggregate inflation expectations

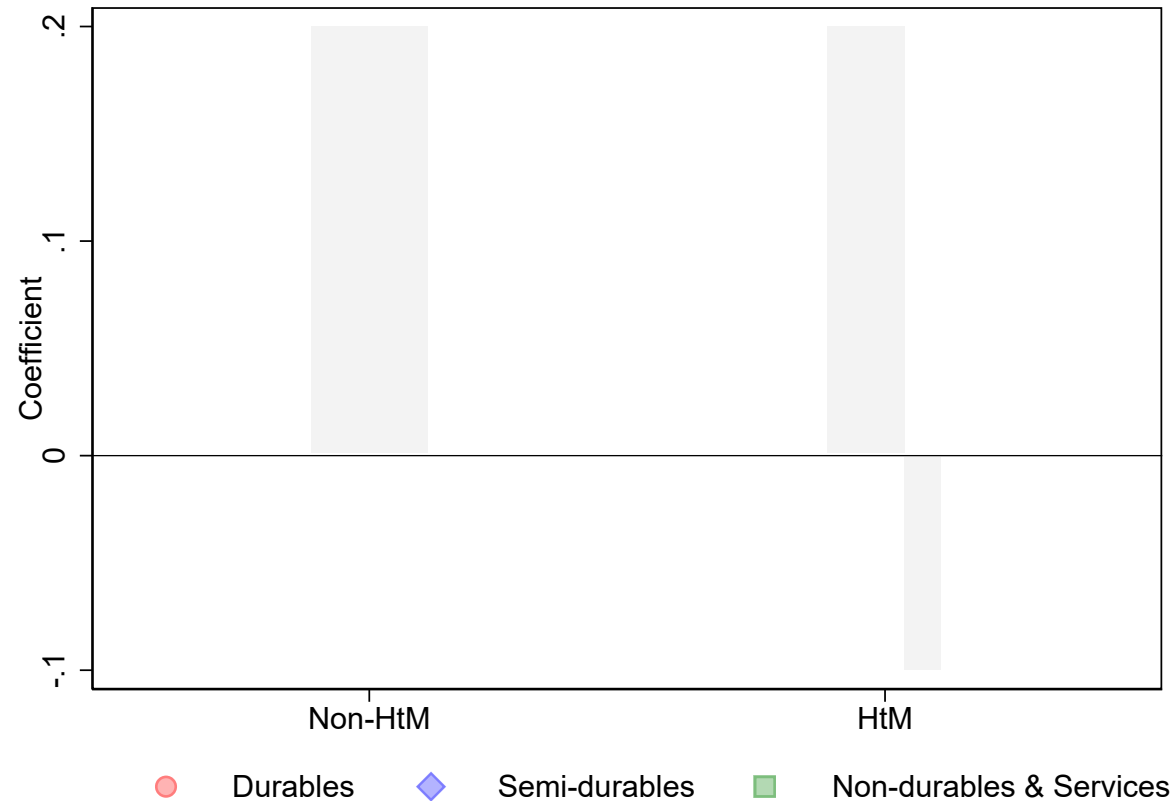
Consumption responses to higher aggregate inflation expectations by HtM status



- **Non-HtM** → increases spending in **all consumption categories**
- **HtM** → increases **durable investment**, sacrificing non-durables (**reallocation**)

Evidence | Higher aggregate inflation expectations, pooled

Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$

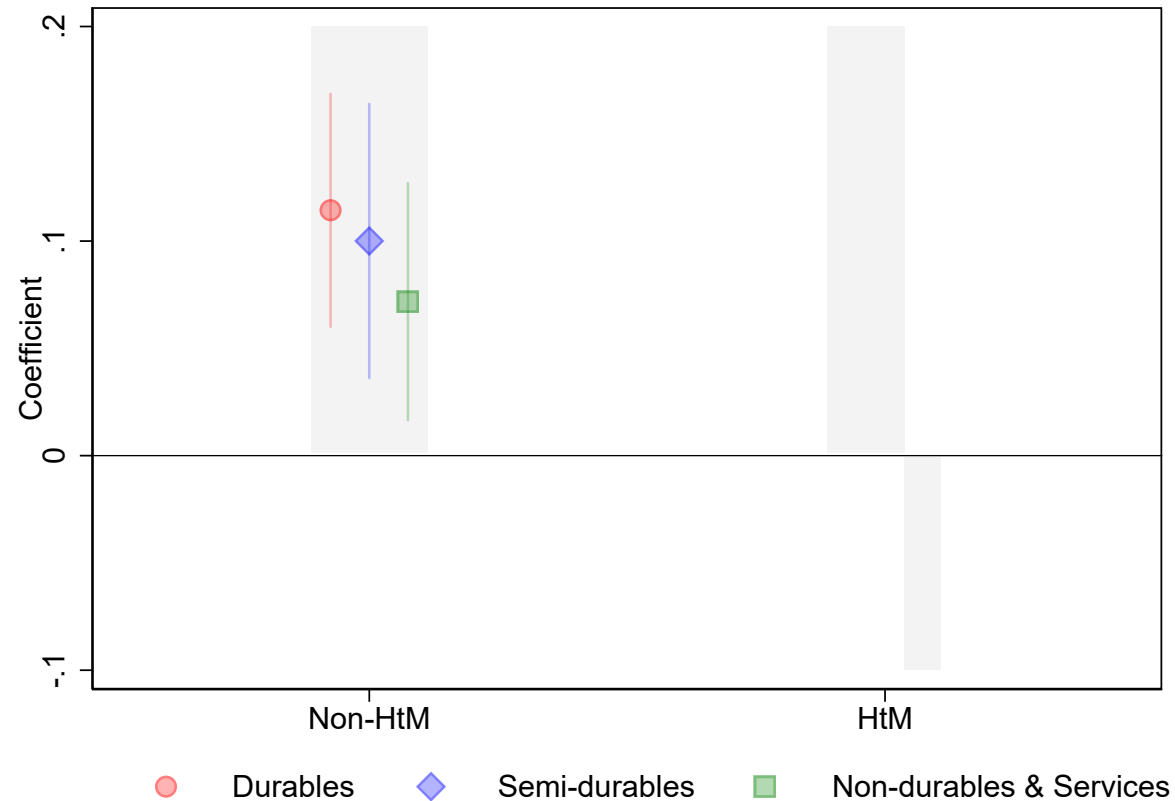


► Full results

► Sensitivity

Evidence | Higher aggregate inflation expectations, pooled

Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



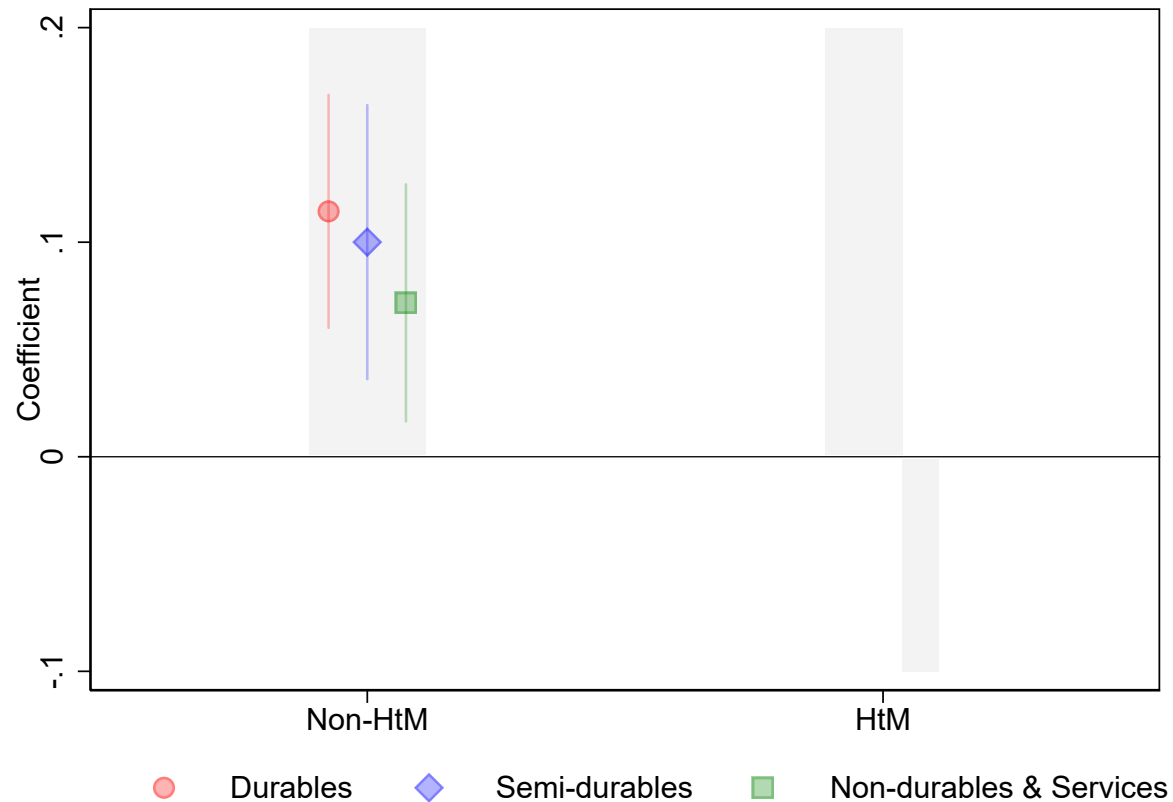
- **Non-HtM** $\rightarrow$ increases spending in **all consumption categories**

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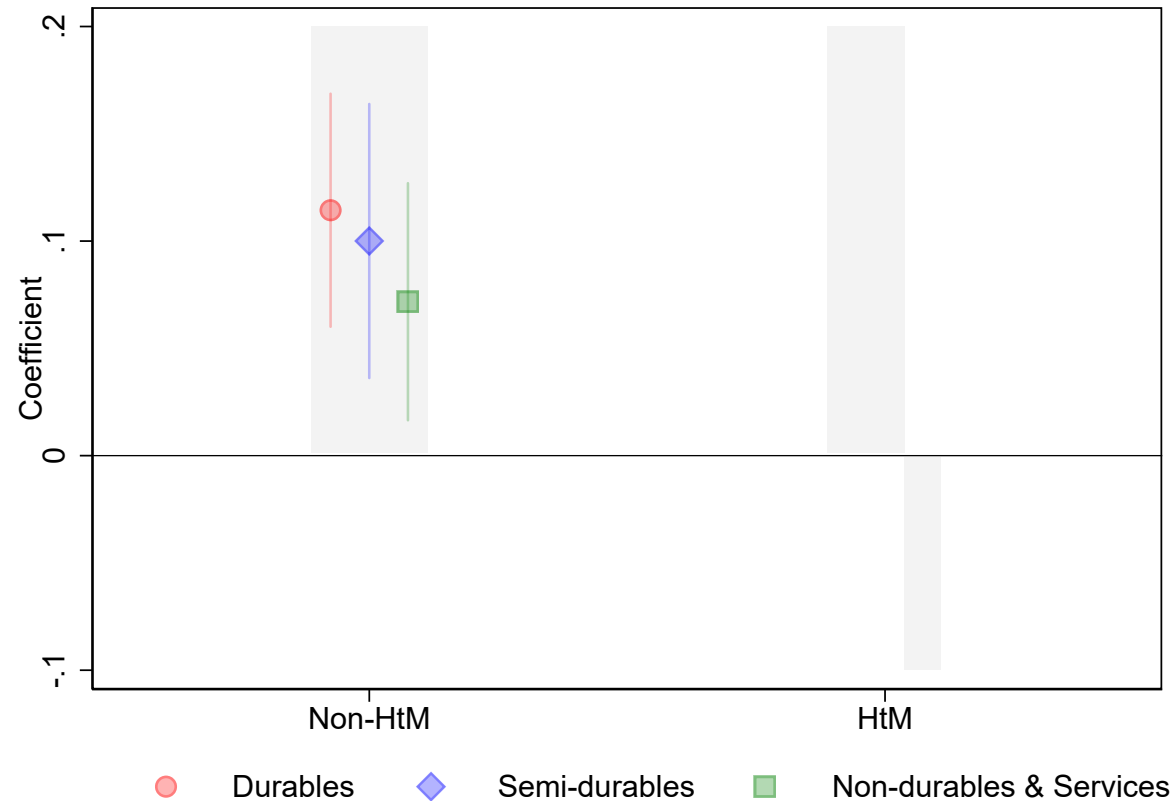
Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



- $\uparrow 1$ p.p. in non-HtM aggregate inflation expectations...
 $\Rightarrow \uparrow \approx 0.1$ p.p. in all categories & $\uparrow \approx 1.1$ p.p. annually in overall consumption

Evidence | Higher aggregate inflation expectations, pooled

Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



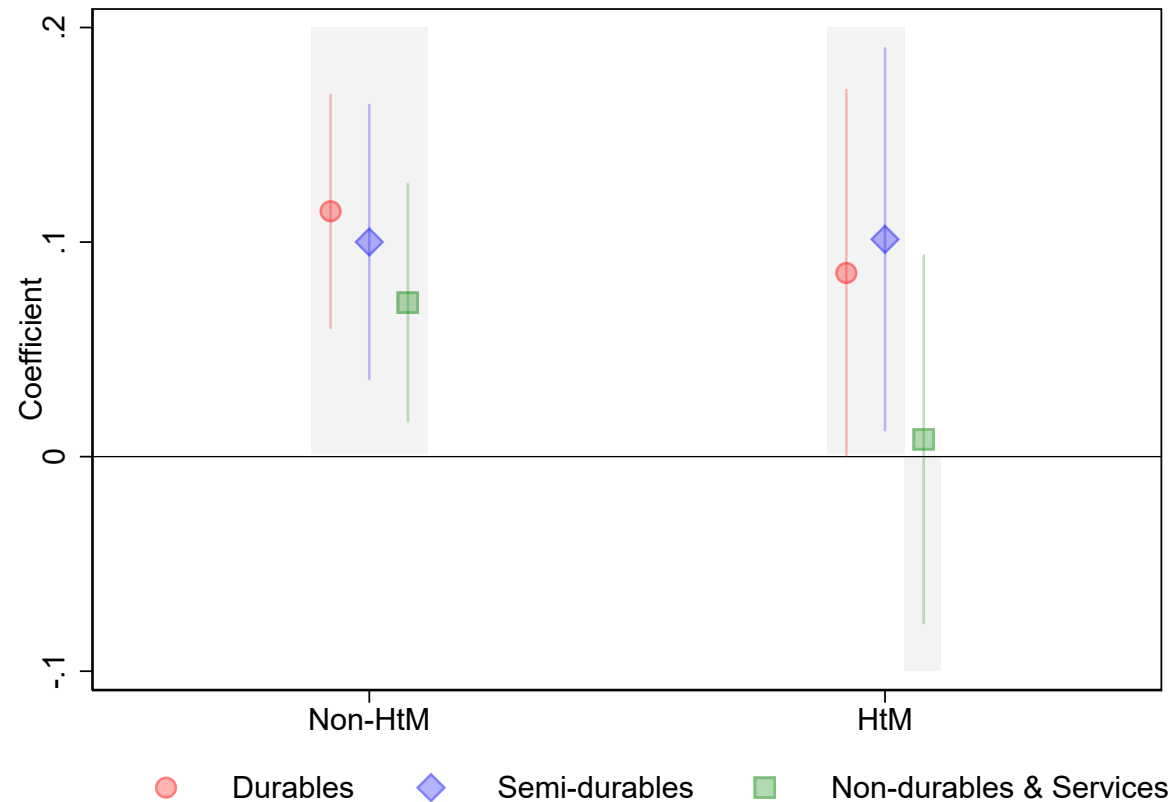
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Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



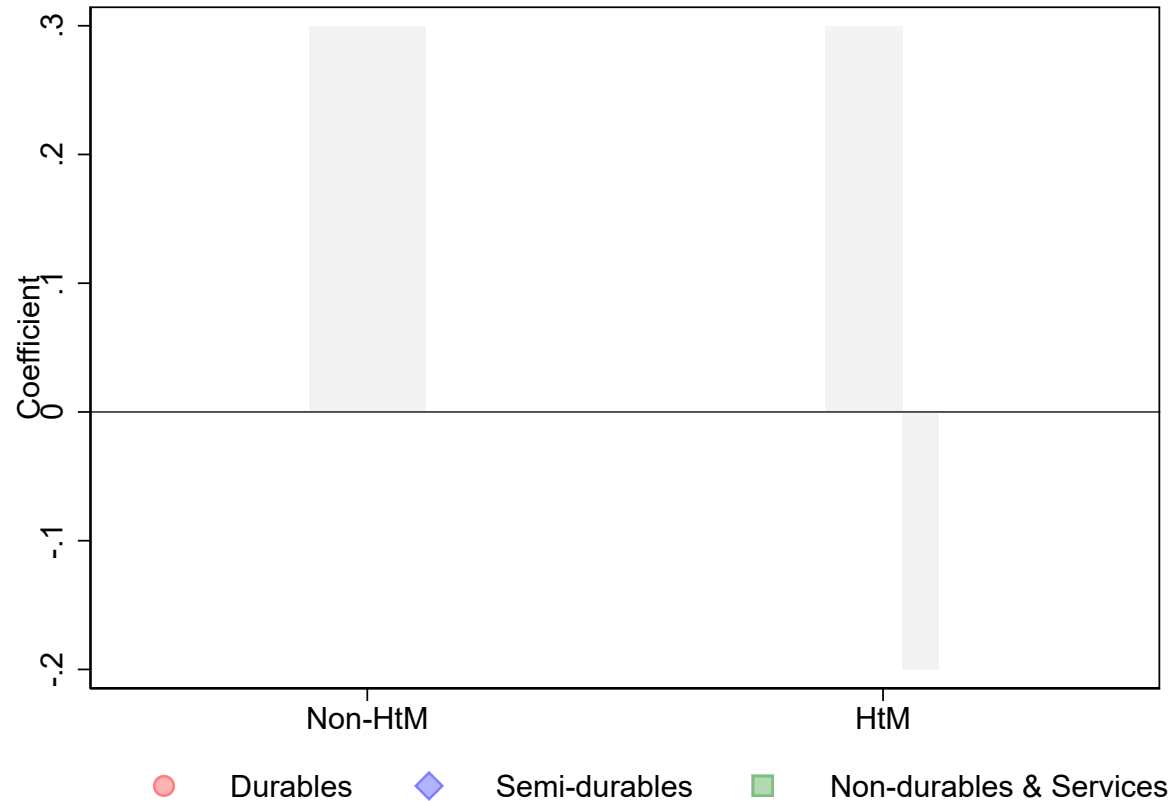
- **Non-HtM** $\rightarrow$ increases spending in **all consumption categories**
- **HtM** $\rightarrow$ selectively **increases durable investment**, keeping non-durables flat
- Strongly aligned with Prediction 1

► Full results

► Sensitivity

PE | Prediction 2: Higher durable inflation expectations

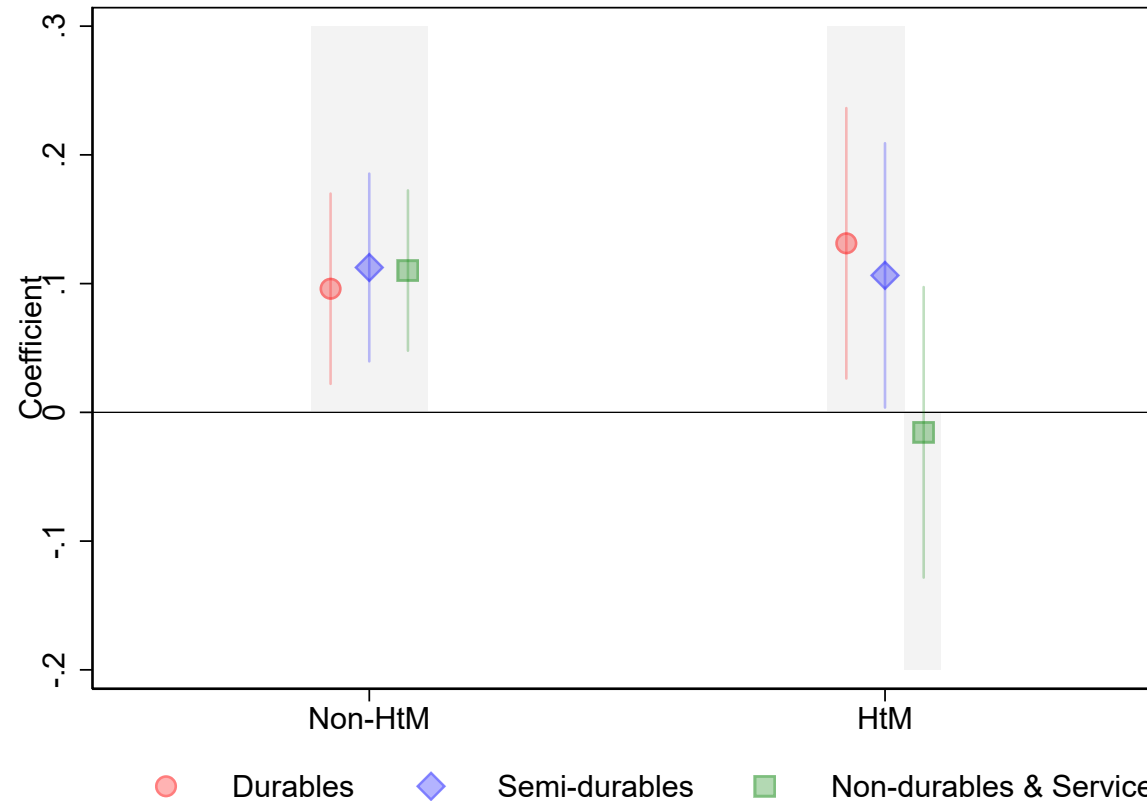
Consumption responses to higher durable-goods inflation expectations by HtM status



- **Non-HtM** → broad-based spending increase
- **HtM** → strong shift toward durables, financed by non-durables (reallocation)

Evidence | Higher durable inflation expectations, T2

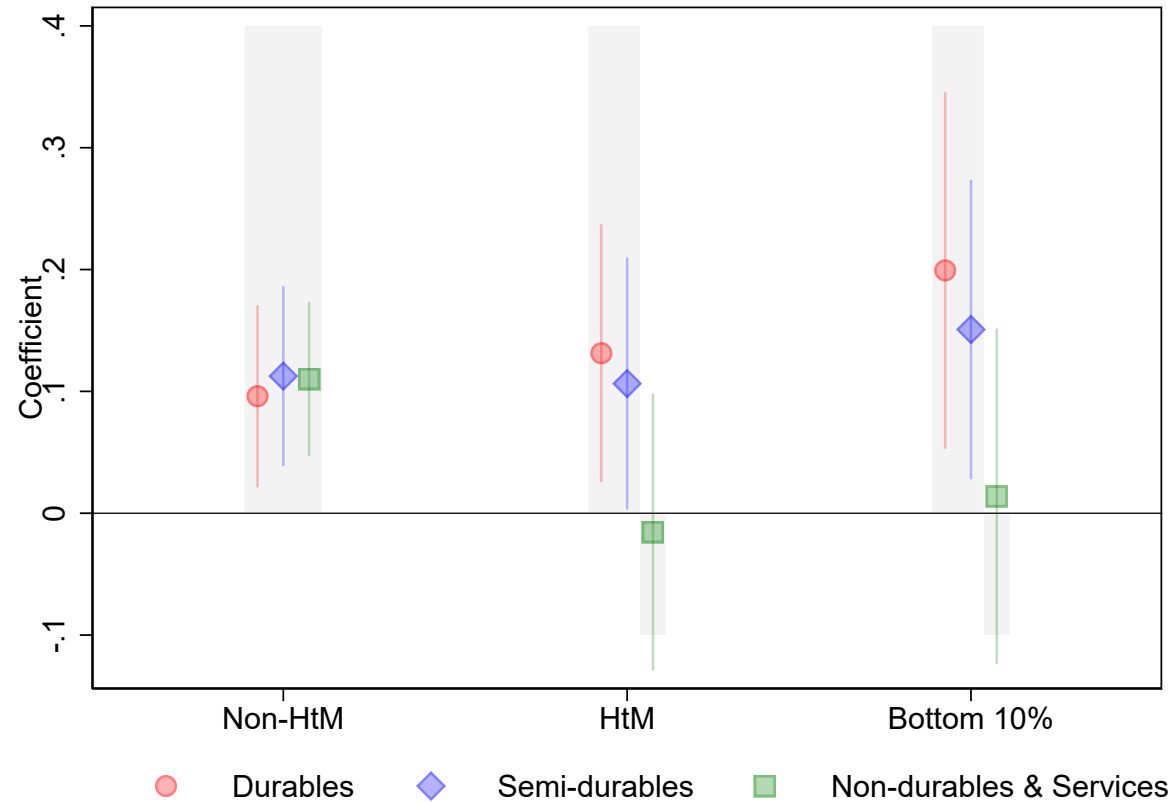
Consumption responses to higher durable-goods inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



- **Non-HtM** $\rightarrow$ broad-based response across consumption categories
- **HtM** $\rightarrow$ durable spending response only, no increase in non-durables
- Strongly consistent with Prediction 2

Evidence | Higher durable inflation expectations, T2

Consumption responses to higher durable-goods inflation expectations of **most restricted HtM**



- **Most restricted HtM: bottom 10%** of PC1 distribution
- Financial constraints **amplify** consumption adjustments
- **Durables as a smoothing mechanism** for most vulnerable households

General equilibrium analysis

Do our PE results matter for GE? | Model sketch

- ▷ **Environment** → textbook TANK model with durables & nominal rigidities
- ▷ Two types of households/HtM & non-HtM consuming non-durables & durables (stock)
- ▷ Rest of the economy
 - Labor-only production of intermediate goods, under nominal rigidities
 - Nominal interest rate set by the monetary authority
 - Abstract from zero-net-supply in the market of bonds
- ▷ **Key mechanism** → Durables introduce a forward-looking margin for HtM in GE

Households | Overview

- ▷ Unit mass of infinitely lived households with identical preferences

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- ▷ Within each type, households are identical → two representative agents: $i = \{H, N\}$
 - HtM, $i = H$: share Λ^H of population
 - non-HtM, $i = N$: share $\Lambda^N = (1 - \Lambda^H)$

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 - HtM, $i = H$: share Λ^H of population
 - non-HtM, $i = N$: share $\Lambda^N = (1 - \Lambda^H)$
- ▷ Different from the PE model
 - Endogenous labor supply → CES aggregator $l_{i,t}$
 - ↪ $l_{i,j,t}$: labor supplied in sector j
 - Only non-HtM holds equity (Bilbiie, 2008)
 - ↪ $\Omega_{j,t-1}$: equity shares in sector j at the beginning of period t
 - ↪ $v_{j,t}$: average market value per share
 - ↪ $P_{j,t}\Pi_{j,t}$: nominal profits per share

Households | Optimization problem

▷ Representative household $i = \{H, N\}$ solves

$$\max_{\{c_{i,k}, z_{i,k}, \{l_{i,j,k}\}_{j=\{c,d\}}, \{B_k, \{\Omega_{j,k}\}_{j=\{d,c\}}\} \cdot \mathbb{1}_{i=N}\}_{k \geq t}} \sum_{k \geq t} \beta^{k-t} \mathbb{E}_t [u(c_{i,k}, d_{i,k}, l_{i,k})]$$

$$\begin{aligned} \text{s.t.} \quad P_{c,t} c_{i,t} + P_{d,t} z_{i,t} &= P_{w,t} \sum_{j=\{d,c\}} l_{i,j,t} + T_{i,t} + (R_{t-1} B_{t-1} - B_t) \cdot \mathbb{1}_{i=N} \\ &+ \sum_{j=\{d,c\}} (\Omega_{j,t-1} (v_{j,t} + P_{j,t} \Pi_{j,t}) - \Omega_{j,t} v_{j,t}) \cdot \mathbb{1}_{i=N} \end{aligned}$$

$$d_{i,t} = z_{i,t} + (1 - \delta) d_{i,t-1}$$

▷ non-HtM transversality condition $\rightarrow \lim_{k \rightarrow \infty} \prod_{h=0}^k (R_{t+h-1})^{-1} B_{t+k} = 0$

► Optimality conditions

► Preferences

Firms

- ▷ Standard two-sector NK production with Calvo pricing: $j = \{c, d\}$
- ▷ Split the production process into two stages
 - A perfectly competitive final good producer aggregates ...
 - ... intermediate varieties supplied by monopolistically competitive firms
- ▷ Labor-only technology
- ▷ NK optimal subsidy → removes steady-state markup distortions (Rotemberg and Woodford, 1997)
- ▷ Sector-specific NK Phillips Curve (Calvo, 1983)

$$\hat{\pi}_{j,t} = \beta \mathbb{E}_t [\hat{\pi}_{j,t+1}] + (1 - \theta_j) (1 - \beta \theta_j) (\theta_j)^{-1} \widehat{mc}_{j,t}$$

- $mc_{j,t}$: real marginal cost of intermediate firms in sector j
- $(1 - \theta_j)$: probability of price setting of intermediate firms in sector j

Monetary policy & savings counterpart

▷ **Policy instrument** → adjusts to sectoral **inflation** & a non-systematic **disturbance**

$$R_t = \bar{R}^{1-\rho_R} R_{t-1}^{\rho_R} e^{\iota_{m,t}} \prod_{j=\{d,c\}} (\pi_{j,t})^{\phi_j}$$

- $\phi_j > 1$: sector-specific **reaction coefficient**
- $\iota_{m,t}$: AR(1) **monetary policy shock**

Monetary policy & savings counterpart

▷ **Policy instrument** → adjusts to sectoral **inflation** & a non-systematic **disturbance**

$$R_t = \bar{R}^{1-\rho_R} R_{t-1}^{\rho_R} e^{\iota_{m,t}} \prod_{j=\{d,c\}} (\pi_{j,t})^{\phi_j}$$

- $\phi_j > 1$: sector-specific **reaction coefficient**
- $\iota_{m,t}$: AR(1) **monetary policy shock**

▷ **After-profit transfers** → equalized across households

Monetary policy & savings counterpart

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▷ Consumes non-durables using **bonds** $B_{x,t}$ → **absorbs non-HtM's net savings**

Monetary policy & savings counterpart

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▷ To correct saving **imbalances**: $R_{x,t} = R_t e^{-\phi_B B_{x,t}}$ (Schmitt-Grohé and Uribe, 2003)

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▷ To correct saving **imbalances**: $R_{x,t} = R_t e^{-\phi_B B_{x,t}}$ (Schmitt-Grohé and Uribe, 2003)

▷ **Key role** → **equilibrium heterogeneity** between **HtM** & **non-HtM**

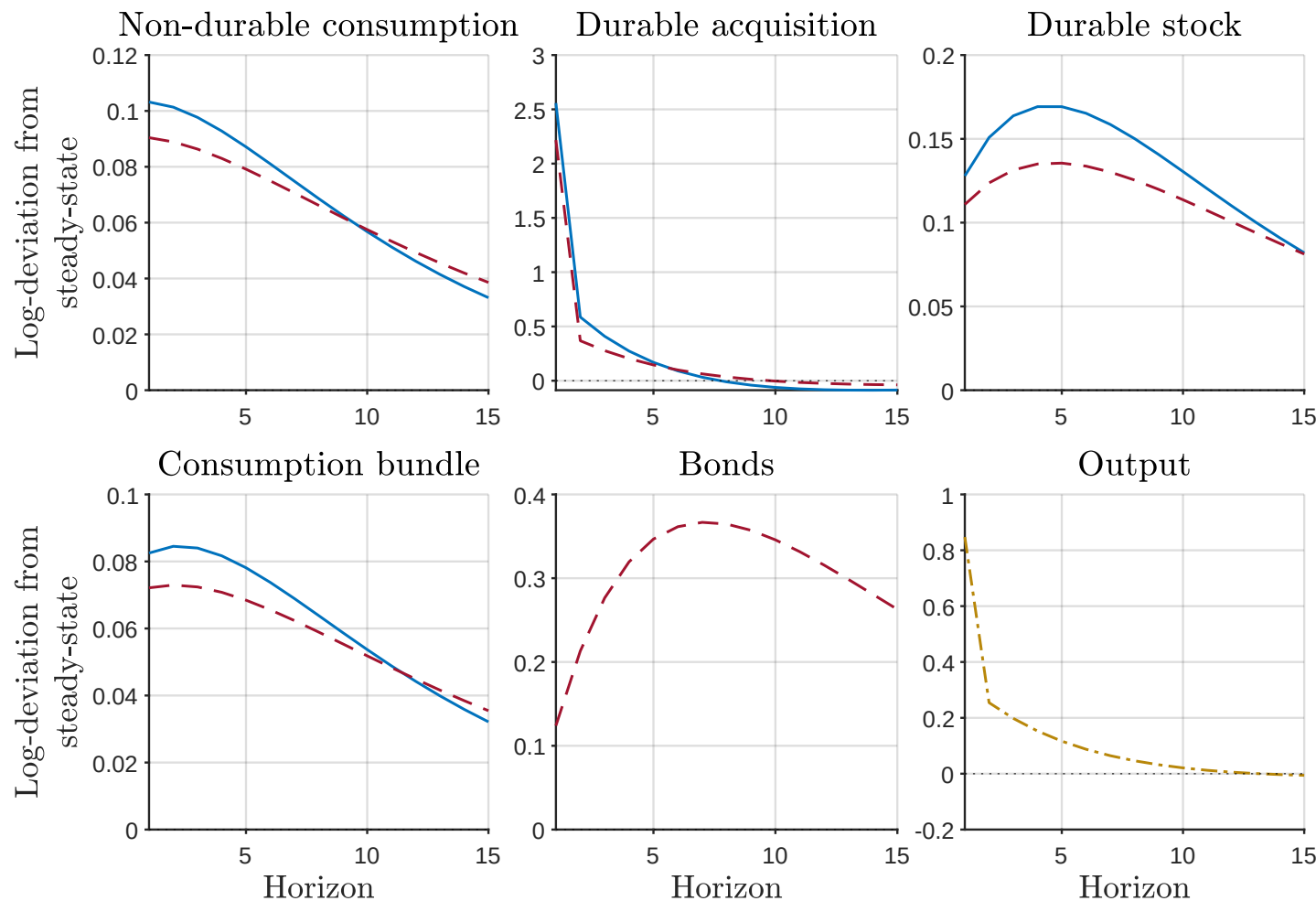
► **Market clearing conditions**

► **Competitive equilibrium**

Calibration

Parameter	Value	Source	Description
β	0.99	Bachmann et al. (2024)	Households' discount factor
σ/ζ	1.00	Bachmann et al. (2024)	Inverse of the intra/intertemporal elasticity of substitution
η	0.50	Bachmann et al. (2024)	Inverse of the Frisch labor supply elasticity
χ	10.0	Bachmann et al. (2024)	Households' relative labor utility cost
δ	0.05	Barsky et al. (2016)	Rate of depreciation of durables
ϕ	0.70	Barsky et al. (2016)	Steady-state share of non-durable consumption
Λ^H	0.40	Aguiar et al. (2025)	Share of HtM households in the population
θ_j	0.75	Bachmann et al. (2024)	Calvo's price stickiness in each sector
ϕ_B	0.10	Schmitt-Grohé and Uribe (2003)	Interest-rate sensitivity to debt
ϕ_j	1.50	Auclert et al. (2020)	Policy reaction to sectoral inflation
ρ_R	0.89	Auclert et al. (2020)	Persistence of monetary policy rate
ρ_m	0.83	Auclert et al. (2020)	Persistence of monetary policy shocks
σ_m	0.57	Auclert et al. (2020)	Standard deviation of monetary policy shocks

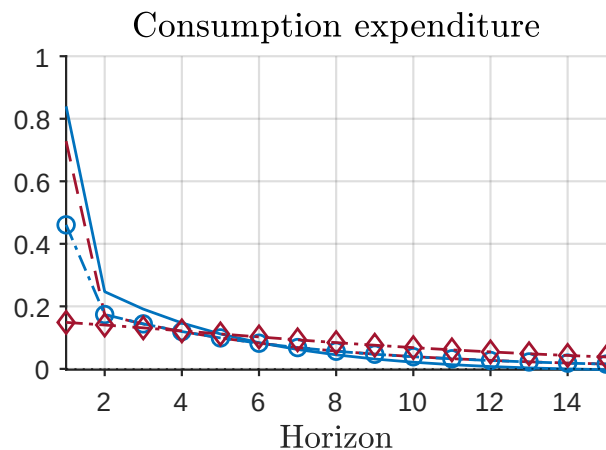
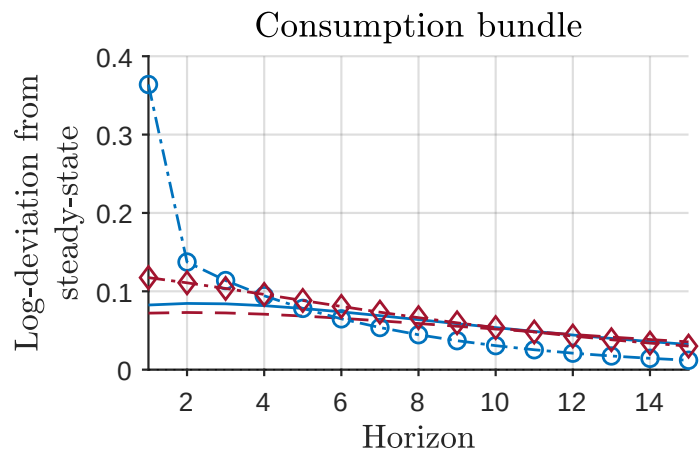
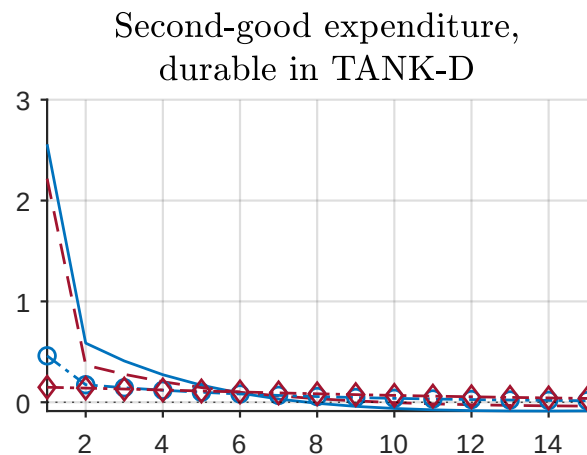
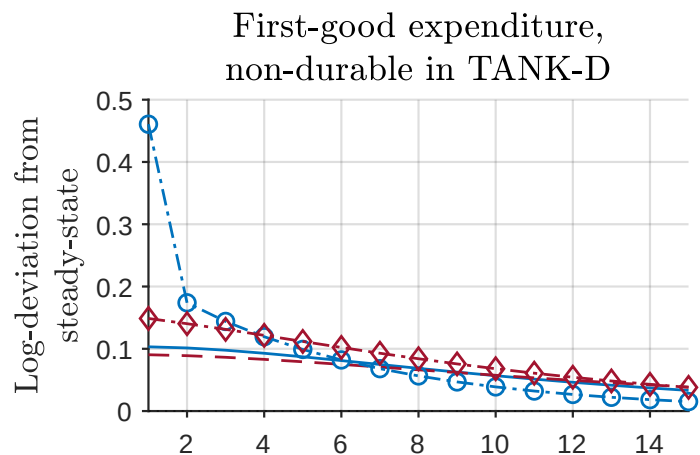
Monetary policy shock | R1: Durables as HtM buffer



- Durables relax **consumption-income** link
- Durables **substitute** for financial assets
- Durable stock gap:
 - More **pronounced & persistent** response
 - Higher income & **strategic consumption**

► All figures

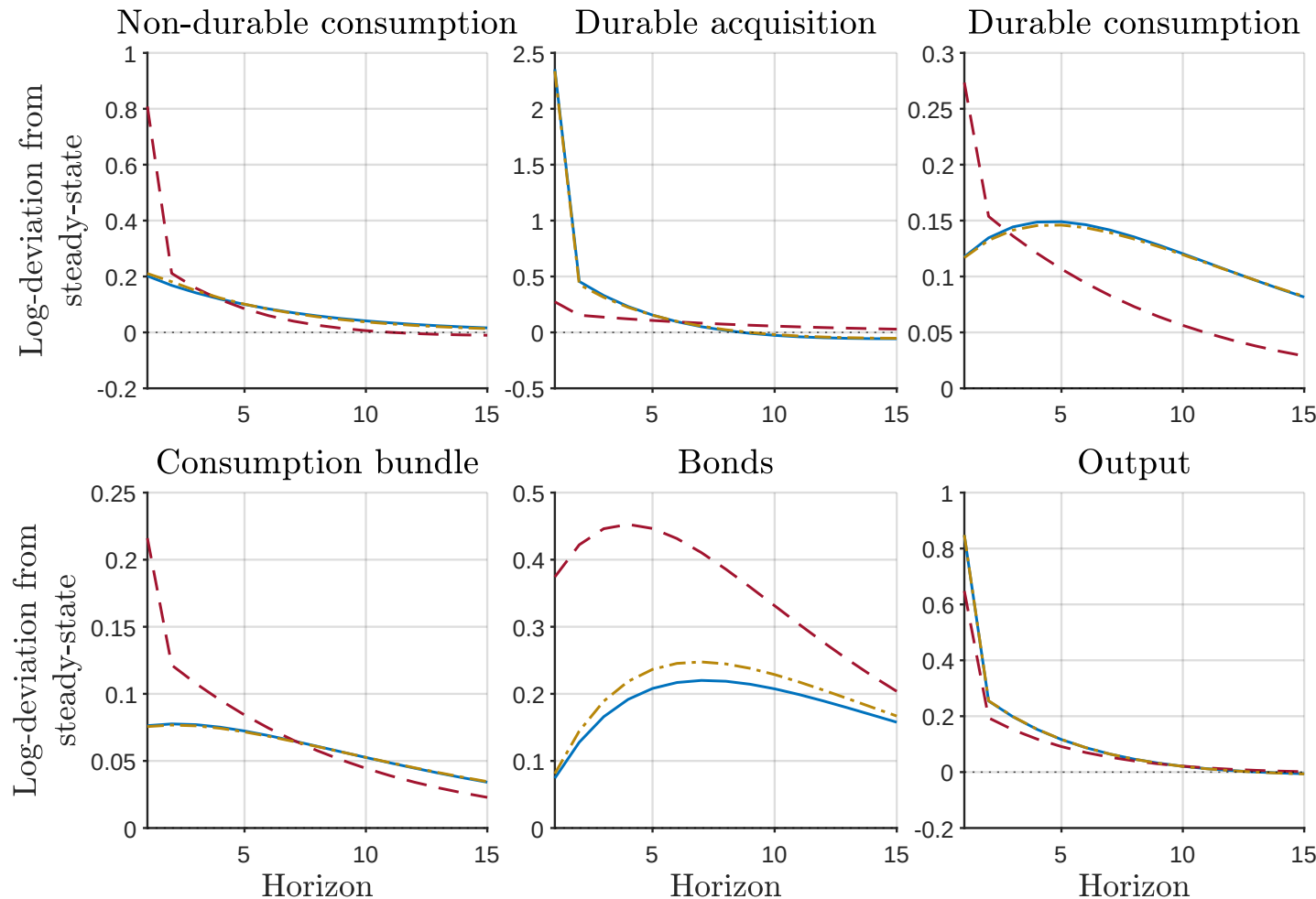
Monetary policy shock | R2: TANK v/s TANK-D



- HtM in TANK-D
- - - non-HtM in TANK-D
- · - · HtM in TANK
- · - · non-HtM in TANK

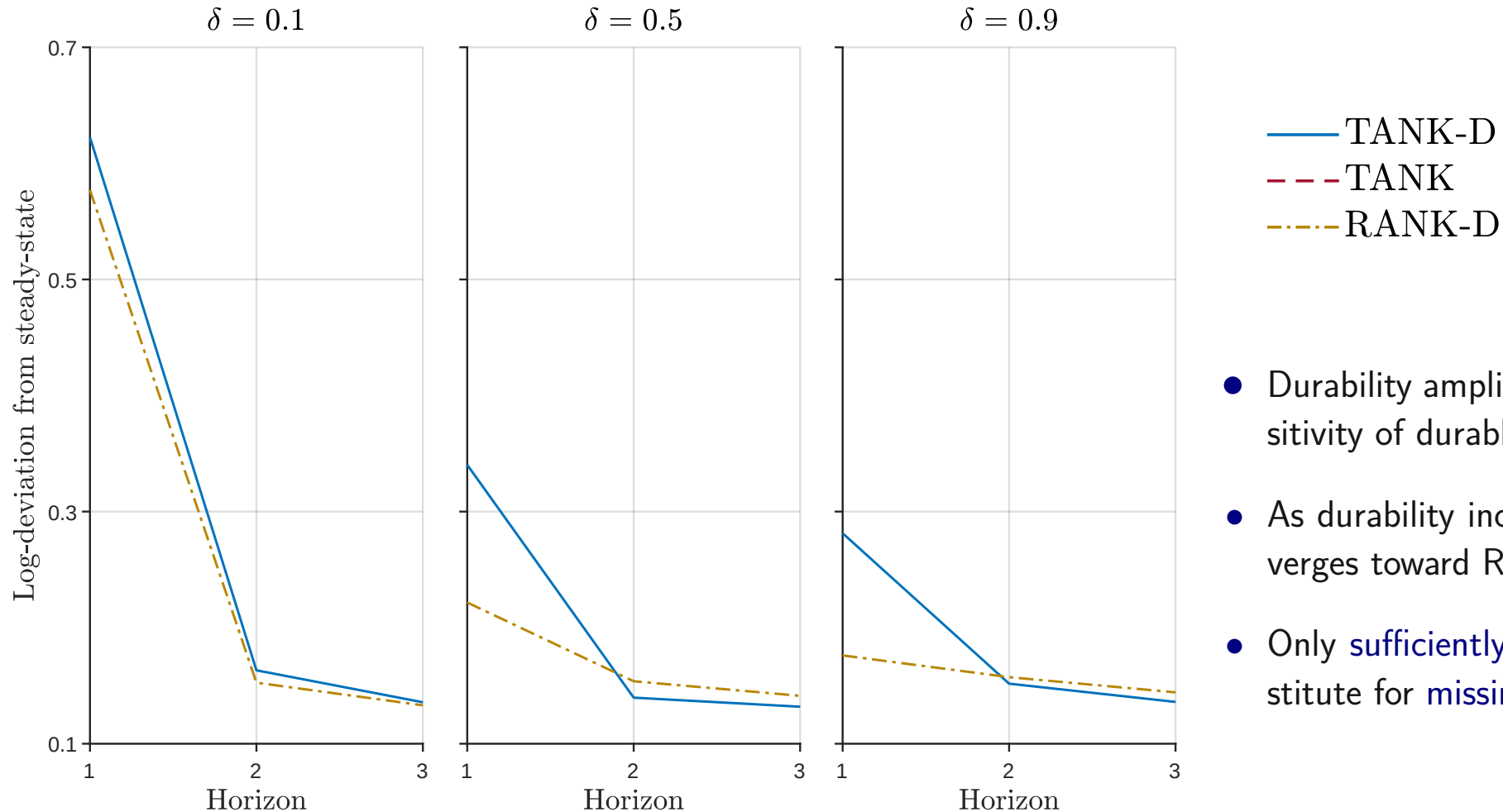
- Financial friction **amplify** on-impact responses
- Durables adjust sharply → **front-loading** to smooth consumption
- Spending becomes **more volatile**... yet consumption is **stabilized**

Monetary policy shock | R3: TANK-D behaves like RANK-D



- Durables dampen the aggregate effects of financial market exclusion
- Traditional result → HtM amplify demand shocks
- Durables mitigate HtM amplification
- Durables provide a smoothing margin, substituting for missing financial market

Monetary policy shock | R3: Durability governs durables' saving role



- Durability amplifies the interest-rate sensitivity of durables
- As durability increases $\rightarrow$ TANK-D converges toward RANK-D
- Only sufficiently durable goods can substitute for missing financial markets

Conclusions

- ▷ Durables fundamentally change the behavior of constrained households
 - ↳ Introduce a forward-looking dimension for HtM
- ▷ HtM forward-looking behavior & expectations matter for individual & aggregate outcomes
- ▷ Durable in a TANK model → dampens the liquidity heterogeneity for aggregate results
- ▷ Aggregate models ignoring durables may overstate the role of liquidity constraints

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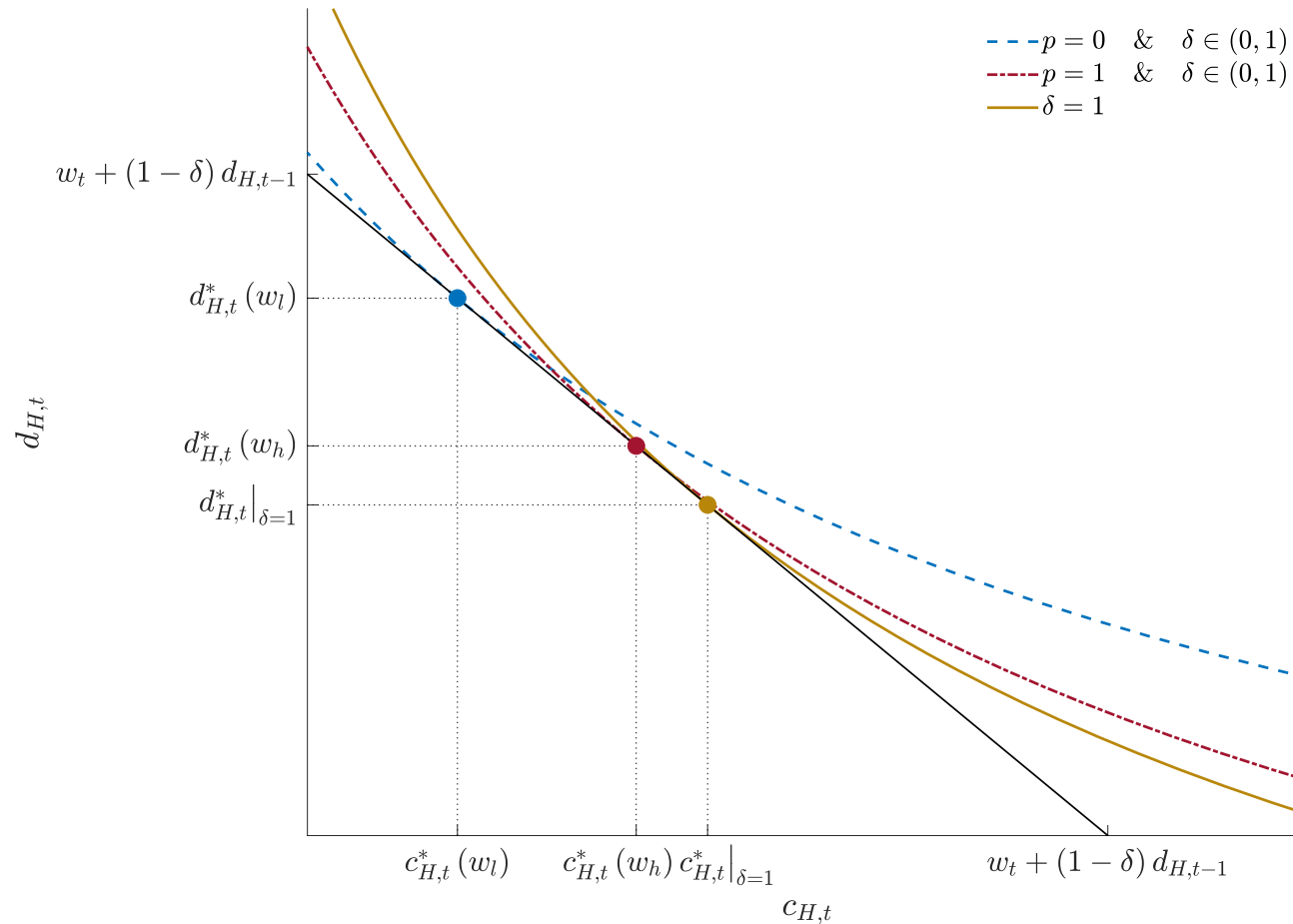
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Optimal HtM consumption



- $\delta = 1 \rightarrow$ allocations independent of future states of the world, p irrelevant
- $\delta \in (0, 1)$ flattens the IC, more when anticipating higher marginal utility

Optimal consumption responses

Lemma 1

Given a **one-time, isolated and exogenous** change in the expected inflation $\mathbb{E}_t [\pi_{j,t+1}]$ of the nominal price $P_{j,t} / j = \{d, c, w\}$, **optimal consumption responses** satisfy

$$\begin{aligned}\frac{\partial z_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} &= \mathcal{D}_i \left(\frac{\partial \mathbb{E}_t [q_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [w_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [r_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \cdot \mathbb{1}_{i=s} \mid \Theta \right) \\ \frac{\partial c_{i,t}}{\partial \mathbb{E}_t [\pi_{j,t+1}]} &= \mathcal{C}_i \left(\frac{\partial \mathbb{E}_t [q_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [w_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]}, \frac{\partial \mathbb{E}_t [r_{t+1}]}{\partial \mathbb{E}_t [\pi_{j,t+1}]} \cdot \mathbb{1}_{i=s} \mid \Theta \right)\end{aligned}$$

where $i = \{H, S\}$, and $\mathcal{D}_i(\bullet), \mathcal{C}_i(\bullet) : \mathbb{R}^3 \rightarrow \mathbb{R}$.

- Responses: hinge on changes in **expected relative prices**
- Policy function: **household-specific**
- HtM: **no sensitivity** to changes in expected **real interest rate**

HtM candidates | Survey questions

Question 1 : *paycheck-to-paycheck*

QH.1 Thinking about your current household's situation, how strongly do you agree or disagree with the statement: "My household lives *paycheck-to-paycheck*"?

- ☐ I strongly agree.
- ☐ I somewhat agree.
- ☐ I neither agree nor disagree.
- ☐ I somewhat disagree.
- ☐ I strongly disagree.

Question 2 : *borrow*

QH.2 If your household needed to *borrow one week's worth of your monthly household income*, how easy or difficult would it be for your household to come up with the money?

- ☐ It would be extremely difficult.
- ☐ It would be relatively difficult.
- ☐ It would be neither easy nor difficult.
- ☐ It would be relatively easy.
- ☐ It would be extremely easy.

HtM candidates | Survey questions

Question 3 : *savings*

QH.3 Which fraction of your income do you [save](#)?

- ☐ I spend all of my income each month.
- ☐ I spend more money than I earn, and I often use credit cards or other loans to supplement my monthly income.
- ☐ I save around 10% of my monthly income.
- ☐ I save around 25% of my monthly income.
- ☐ I save at least 50% of my monthly income.

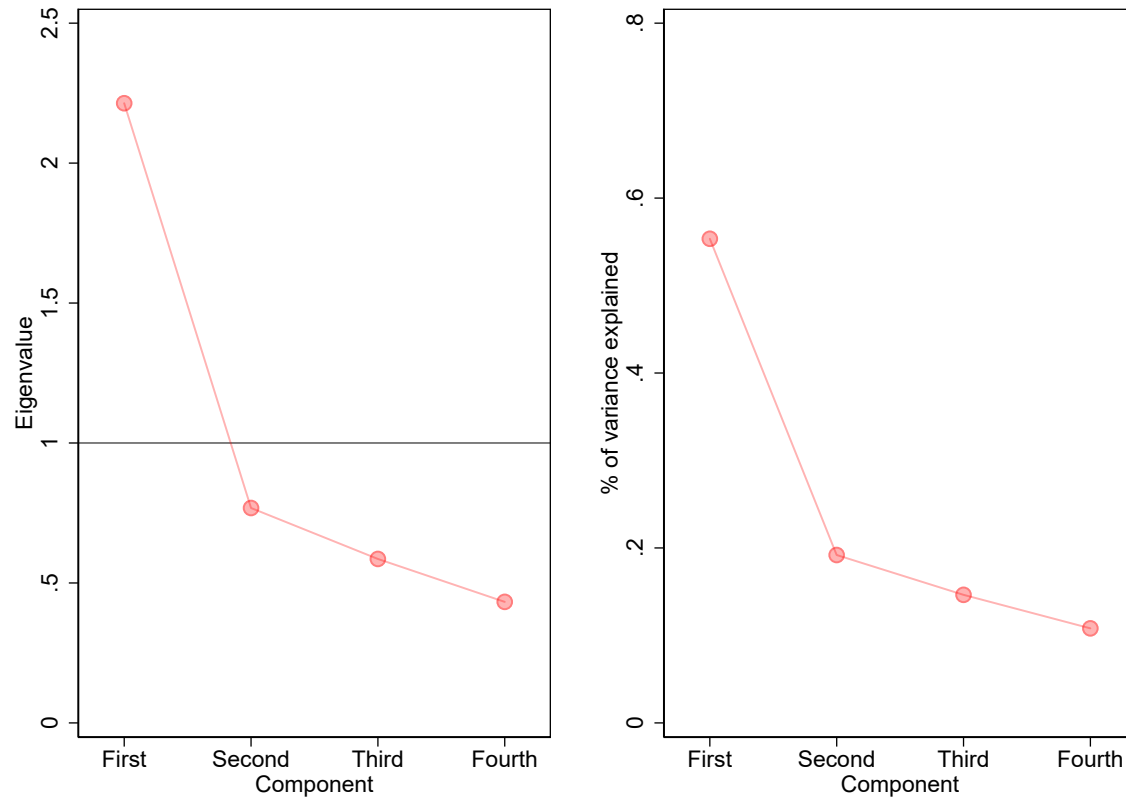
Question 4 : *weeks*

QH.4 Suppose your household loses its primary source of income today. [How long can your household cover its weekly expenses](#) with your household's total money in checking and savings accounts or other liquid financial assets such as cash, stocks, government and corporate bonds, and mutual and money market funds? Please give your best guess.

- If my household loses its primary source of income today, we can cover our expenses for _____ week(s) using the money in our checking and saving accounts plus other liquid financial assets.

Dominant & significant share of common variance

Eigenvalue & variance explained by each principal component



Balanced sample across treatment arms

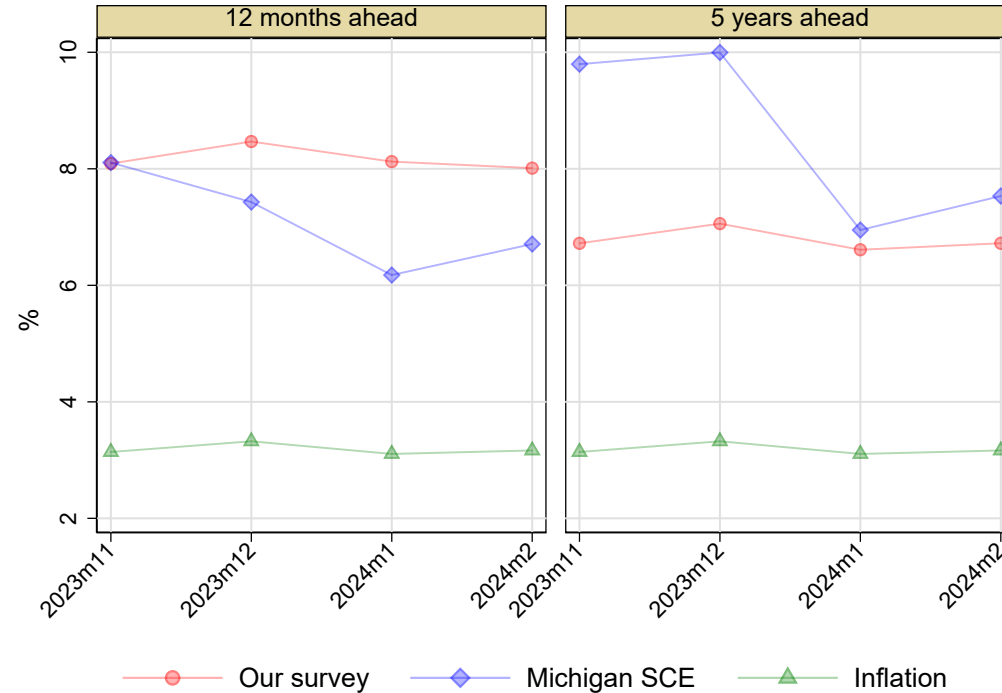
Balance test on pre-treatment covariates

	T-test or χ^2 -test p-value					
	C & T1	C & T2	C & P	T1 & T2	T1 & P	T2 & P
Age	0.92	0.82	0.23	0.92	0.34	0.40
Gender	0.54	0.37	0.99	0.81	0.59	0.44
Education level	0.55	0.14	0.82	0.13	0.39	0.75
Civil status	0.22	0.20	0.24	0.70	0.34	0.64
Household net income	0.09	0.32	0.07	0.05	0.10	0.05
Number of children	0.58	0.91	0.67	0.70	0.94	0.77
Political affiliation	0.51	0.95	0.74	0.27	0.77	0.45
English fluency	0.46	0.08	0.39	0.56	0.21	0.05

- T-tests or χ^2 -tests used depending on the nature of the variable
- The second row signal the comparison groups in each column
- No significant differences across treatment arms

Prior & posterior beliefs

Inflation expectations and actual inflation rate



Note: The figure presents the time series of the averages of our collected data and the raw data from the University of Michigan's Survey of Consumers. The series plotted are the annual inflation expectations 12 months and 5 years ahead, alongside the actual inflation rate. Our collected inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

Descriptive statistics by survey & HtM status

Summary statistics on respondents' inflation expectations

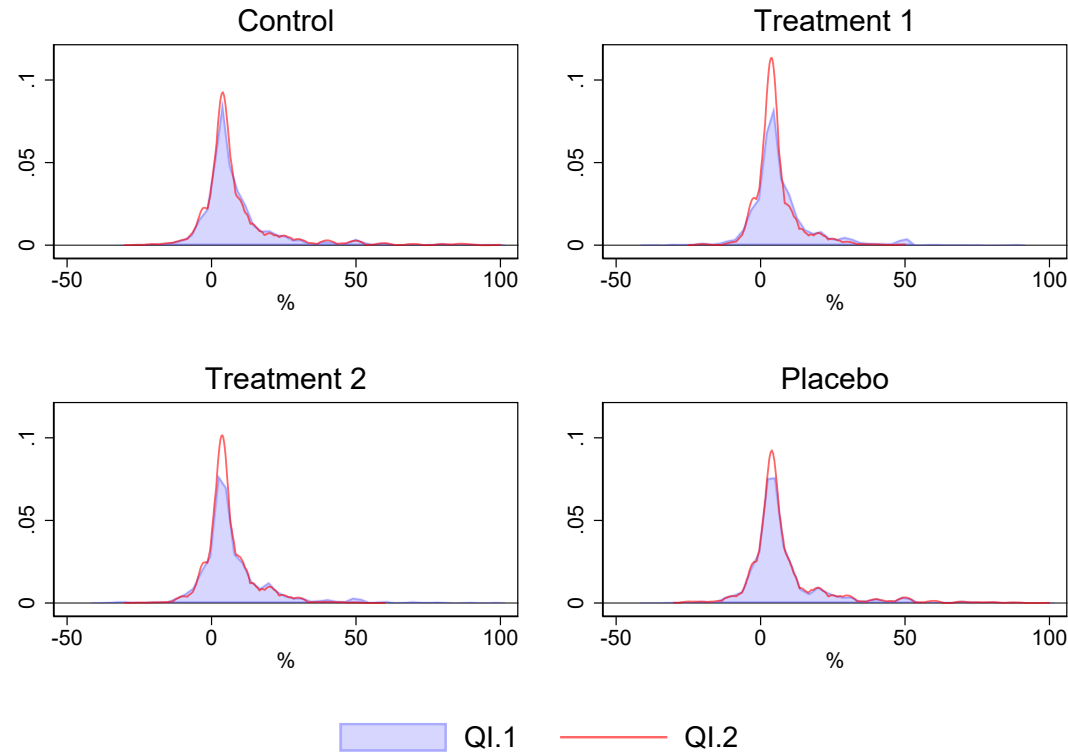
Inflation expectations	Michigan SCE		Full sample		HtM		non-HtM	
	Mean	Median	Mean	Median	Mean	Median	Mean	Median
12 months ahead	6.8	6.7	8.2	5.0	8.7	5.0	7.8	4.0
5 years ahead	8.3	7.5	6.8	4.0	6.9	5.0	6.6	4.0

Note: The table reports descriptive statistics of the answers to the prior (Q1.1) and posterior (Q1.2) questions using our collected data and the raw data from the University of Michigan's Survey of Consumers. Our collected inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

[Return](#)

Treatments | Beliefs updating

Kernel density of inflation expectations by treatment status



- T1 & T2 posterior densities → **bunching** at the treatment signal & **higher kurtosis**
- **Successful treatments** affecting beliefs, suggestive evidence

Treatments | Validation

Treatment effects on medium-run inflation expectations

Variable _j		Variable _j	Variable _j × H _j
<i>prior_j</i>	0.856*** (0.012)	0.868*** (0.013)	−0.033 (0.023)
<i>treat_{j,1}</i>	1.798*** (0.197)	1.665*** (0.224)	0.284 (0.289)
<i>treat_{j,1} × prior_j</i>	− 0.486*** (0.018)	− 0.465*** (0.023)	−0.041 (0.037)
<i>treat_{i,2}</i>	1.894*** (0.204)	1.553*** (0.226)	0.926*** (0.324)
<i>treat_{j,2} × prior_j</i>	− 0.459*** (0.017)	− 0.455*** (0.021)	−0.010 (0.035)
<i>treat_{i,3}</i>	0.149 (0.194)	−0.036 (0.221)	0.424 (0.278)
<i>treat_{j,3} × prior_j</i>	0.001 (0.016)	0.000 (0.019)	0.005 (0.035)
Fixed-effects	Yes	Yes	
Controls	Yes	Yes	
R ²	0.792	0.793	
Observations	5,529	5,529	

Note: The set of fixed effects includes week and U.S. state fixed effects. The inflation expectations data have been trimmed at the weekly frequency by excluding the top and bottom 2.5% of the distribution.

Heterogeneous effects by HtM status | 2SLS

▷ First-stage:

$$\text{post}_j \times (1 - H_j) = \alpha^N + \mathbf{X}_j (\boldsymbol{\beta}^N)' + \eta^N \cdot \text{prior}_j + \sum_{\tau=\{1,2\}} \left[\gamma_{\tau,1}^N (\text{treat}_{j,\tau} \times (1 - H_j)) + \gamma_{\tau,2}^N (\text{treat}_{j,\tau} \times \text{prior}_j \times (1 - H_j)) \right] + v_j^N$$

$$\text{post}_j \times H_j = \alpha^H + \mathbf{X}_j (\boldsymbol{\beta}^H)' + \eta^H \cdot \text{prior}_j + \sum_{\tau=\{1,2\}} \left[\gamma_{\tau,1}^H (\text{treat}_{j,\tau} \times H_j) + \gamma_{\tau,2}^H (\text{treat}_{j,\tau} \times \text{prior}_j \times H_j) \right] + v_j^H$$

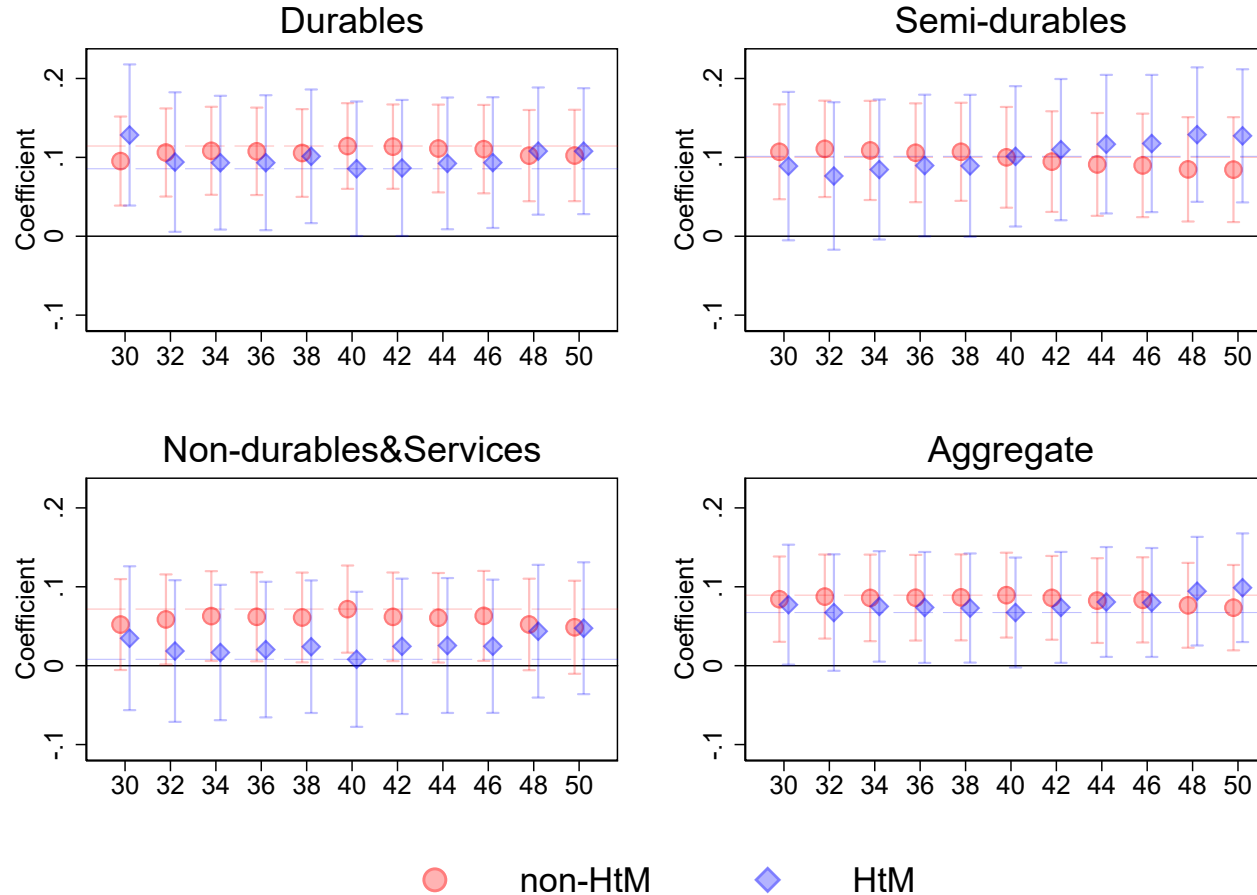
▷ Second-stage:

$$\% \Delta c_{j,k,h} = \alpha_{k,h} + \mathbf{X}_j \boldsymbol{\beta}'_{k,h} + \eta_{k,h} \cdot \text{prior}_j + \theta_{k,h}^N \cdot \overbrace{\text{post}_j \times (1 - H_j)} + \theta_{k,h}^H \cdot \overbrace{\text{post}_j \times H_j} + \varepsilon_{j,k,h}$$

Return

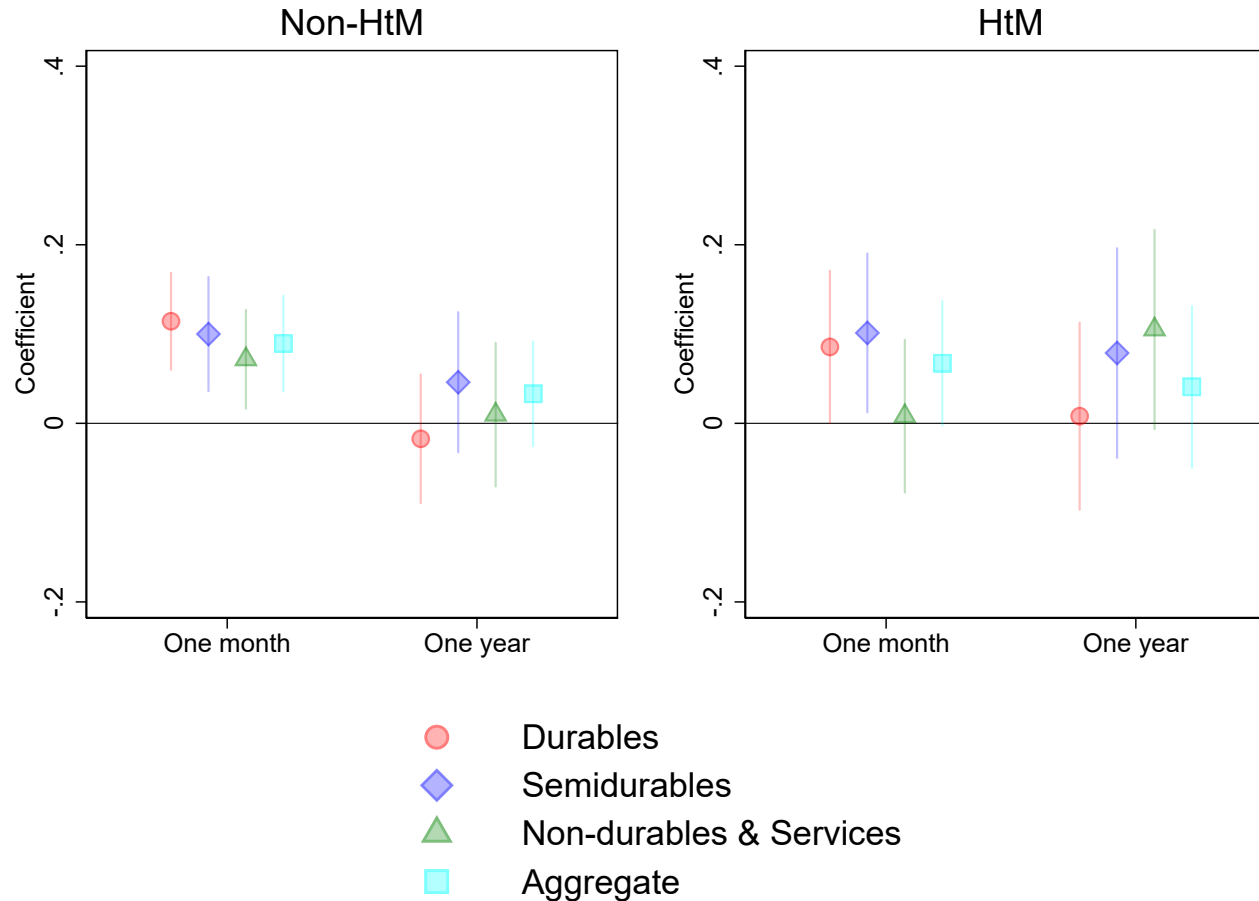
Stability of the inflation expectations coefficients

Estimated coefficients with alternative HtM cutoff



Evidence | Higher aggregate inflation expectations, pooled

Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



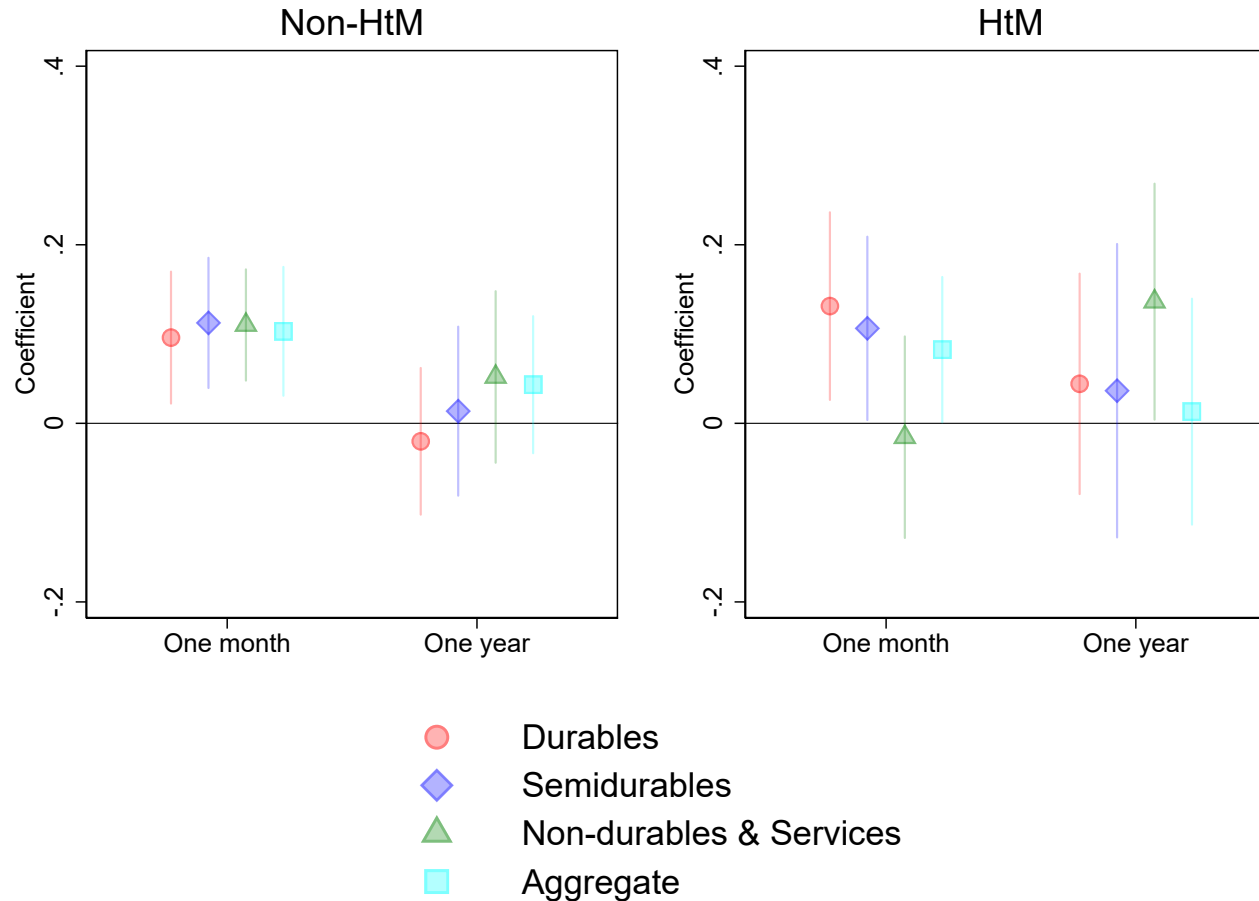
Evidence | Higher aggregate inflation expectations, pooled

Consumption responses to higher aggregate inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$

	% $\Delta c_{j,k,h}^e$: Intended consumption growth on category k at horizon h							
	Durables		Semi-durables		Non-durables and Services		Aggregate	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$post_j \times (1 - H_j)$	0.114*** (0.033)	-0.017 (0.043)	0.100*** (0.038)	0.046 (0.047)	0.071** (0.033)	0.009 (0.048)	0.089*** (0.032)	0.033 (0.035)
$post_j \times H_j$	0.085* (0.051)	0.008 (0.063)	0.101* (0.054)	0.078 (0.071)	0.008 (0.052)	0.105 (0.067)	0.067 (0.042)	0.040 (0.055)
Horizon	One month	One year	One month	One year	One month	One year	One month	One year
Observations	5,099	5,087	5,095	5,106	4,946	4,953	5,147	5,160
Fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat, non-HtM	179.9	176.9	174.0	178.5	171.6	173.7	183.5	184.2
F-stat, HtM	154.8	157.4	158.5	159.0	145.1	151.3	154.0	155.9
R ²	0.134	0.069	0.139	0.079	0.150	0.077	0.223	0.205

Evidence | Higher durable inflation expectations, T2

Consumption responses to higher durable-goods inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$



Evidence | Higher durable inflation expectations, T2

Consumption responses to higher durable-goods inflation expectations by HtM status $\rightarrow \theta_{k,h}^i$

	% $\Delta c_{j,k,h}^e$: Intended consumption growth on category k at horizon h							
	Durables		Semi-durables		Non-durables and Services		Aggregate	
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
$post_j \times (1 - H_j)$	0.096** (0.044)	-0.020 (0.049)	0.112** (0.044)	0.013 (0.057)	0.110** (0.037)	0.052 (0.058)	0.103** (0.043)	0.043 (0.046)
$post_j \times H_j$	0.131** (0.063)	0.044 (0.075)	0.106* (0.062)	0.036 (0.099)	-0.015 (0.037)	0.136* (0.058)	0.082* (0.049)	0.013 (0.076)
Horizon	One month	One year	One month	One year	One month	One year	One month	One year
Observations	3,788	3,776	3,796	3,810	3,689	3,690	3,837	3,843
Fixed-effects	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
F-stat, non-HtM	190.0	186.5	188.3	201.1	198.2	197.1	208.3	203.7
F-stat, HtM	164.4	164.4	167.7	169.7	155.9	163.5	163.1	166.5
R ²	0.146	0.074	0.153	0.093	0.161	0.094	0.224	0.204

Households | Optimality conditions

▷ Conditions common to HtM & non-HtM, $i = \{H, N\}$

$$\text{Labor supply condition : } w_t = - \frac{mu_{i,t}^l}{mu_{i,t}^c} \frac{\partial l_{i,t}}{\partial l_{i,j,t}}$$

$$\text{Durables Euler equation : } mu_{i,t}^d = q_t mu_{i,t}^c \left(1 + \frac{\partial \psi_{i,t}}{\partial d_{i,t}} \right) - \beta \mathbb{E}_t \left[q_{t+1} mu_{i,t+1}^c \left((1 - \delta) \left(1 + \frac{\partial \psi_{i,t+1}}{\partial d_{i,t+1}} \right) - \frac{\partial \psi_{i,t+1}}{\partial d_{i,t}} \right) \right]$$

▷ Additional conditions for non-HtM, $i = N$

$$\text{Bond Euler equation : } 1 = \beta R_{t+1} \mathbb{E}_t \left[mu_{N,t+1}^c (mu_{N,t}^c)^{-1} (\pi_{c,t+1})^{-1} \right]$$

$$\text{Equity Euler equation : } v_{j,t} = \beta \mathbb{E}_t \left[mu_{N,t+1}^c (mu_{N,t}^c)^{-1} (v_{j,t+1} + P_{j,t+1} \Pi_{j,t+1}) (\pi_{c,t+1})^{-1} \right]$$

Households | Preferences

▷ Quadratic adjustment costs $\rightarrow \kappa \geq 0$ governs convexity

$$\psi(d_{i,t}; d_{i,t-1}) = \frac{\kappa}{2} \left(\frac{d_{i,t}}{d_{i,t-1}} - 1 \right)^2 d_{i,t}$$

▷ King-Plosser-Rebelo preferences (King et al., 1988)

$$x(c_{i,t}, d_{i,t}) = \left[\varrho^\sigma (c_{i,t})^{1-\sigma} + (1 - \varrho)^\sigma (d_{i,t})^{1-\sigma} \right]^{\frac{1}{1-\sigma}}$$
$$u(x_{i,t}, l_{i,t}) = \frac{(x_{i,t})^{1-\zeta}}{(1-\zeta)} - \chi \frac{(l_{i,t})^{1+\eta}}{(1+\eta)}$$

- $\sigma \in (0, 1)$: inverse of the intratemporal elasticity of substitution
- $\zeta > 0$: inverse of the intertemporal elasticity of substitution
- $\eta > 0$: inverse of the Frisch labor supply elasticity
- $\varrho \in (0, 1)$: preference parameter for non-durable goods

Market clearing conditions

▷ Bonds market clears: $\Lambda^N B_t + B_{x,t} = 0$

▷ Equity market clears in each sector j : $\Omega_{j,t} = (\Lambda^N)^{-1}$

▷ Labor market clears in each sector j :

$$\sum_{i=\{H,N\}} \Lambda^i l_{i,j,t} = \int_{s_j \in \mathcal{F}_j} n_{s_j,t} ds_j = n_{j,t}$$

▷ Both goods markets clear:

$$y_{c,t} = \sum_{i=\{H,N\}} \Lambda^i c_{i,t} + (c_{x,t} - \omega_c)$$
$$y_{d,t} = \sum_{i=\{H,N\}} \Lambda^i [z_{i,t} + \psi(d_{i,t}; d_{i,t-1})]$$

Competitive equilibrium

Given

- Households distribution $\{\Lambda^i\}_{i=\{H,N\}}$
- Stochastic processes for the **monetary policy** $v_{m,t}$ and **productivity shocks** $\{v_{j,t}\}_{j=\{d,c\}}$
- Initial **endogenous state variables** $\{\{d_{i,t-1}, \{B_{t-1}, \{\Omega_{j,t}\}_{j=\{d,c\}}\} \cdot \mathbb{1}_{i=N}\}_{i=\{H,N\}}, B_{x,t-1}\}$

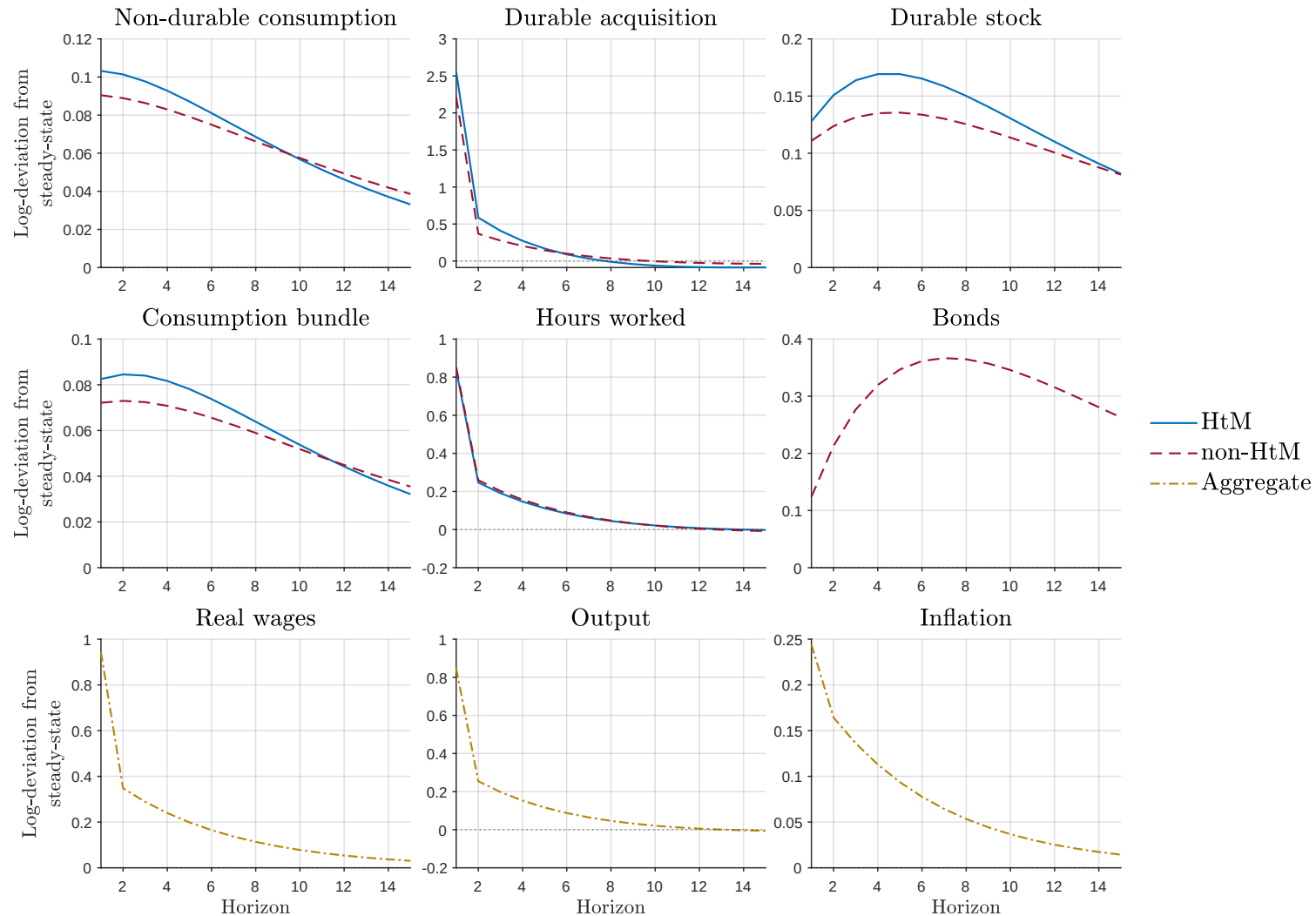
A **competitive equilibrium** is a stochastic sequence of...

- Households and firms decision variables $\{\{c_{i,t}, z_{i,t}, d_{i,t}, \{l_{i,j,t}\}_{j=\{d,c\}}\}_{i=\{H,N\}}, B_t, c_{x,t}, B_{x,t}, \{\Omega_{j,t}, n_{j,t}\}_{j=\{d,c\}}\}$
- Goods and assets prices $\{P_{c,t}, P_{d,t}, P_{w,t}, R_t, \{v_{j,t}\}_{j=\{d,c\}}\}$
- Fiscal policy instruments $\{\{T_{i,t}\}_{i=\{H,N\}}, \{\tau_{j,t}, \nu_{j,t}\}_{j=\{d,c\}}\}$

...such that, at every period t ...

- Households and firms solve their **optimization problem**
- The **monetary policy rule** is satisfied
- All markets clear

Expansionary monetary policy shock | HtM & non-HtM



Expansionary monetary policy shock | TANK-D & RANK-D

